Linköping University | Department of Computer and Information Science
Bachelor's thesis, 16 HP | Datateknik
2026 | LIU-IDA/LITH-EX-G--26/036--SE

Beyond the spreadsheet: Visualizing Patient Flow for Improved Situation Awareness in Emergency Departments

Jonas Hautamäki Bengtsson
Samuel Nikanorsson Sundmark

Supervisor: Jens Wretborn
Supervisor: Klas Arvidsson
Examiner: Emma Enocksson Svensson

LINKÖPING UNIVERSITY

Linköpings universitet
SE–581 83 Linköping
+46 13 28 10 00 , www.liu.se

Upphovsrätt

Detta dokument hålls tillgängligt på Internet – eller dess framtida ersättare – under 25 år från publiceringsdatum under förutsättning att inga extraordinära omständigheter uppstår.

Tillgång till dokumentet innebär tillstånd för var och en att läsa, ladda ner, skriva ut enstaka kopior för enskilt bruk och att använda det oförändrat för icke-kommersiell forskning och undervisning. Överföring av upphovsrätten vid en senare tidpunkt kan inte upphäva detta tillstånd. All annan användning av dokumentet kräver upphovsmannens medgivande. För att garantera äktheten, säkerheten och tillgängligheten finns lösningar av teknisk och administrativ art.

Upphovsmannens ideella rätt innefattar rätt att bli nämnd som upphovsman i den omfattning som god sed kräver vid användning av dokumentet på ovan beskrivna sätt samt skydd mot att dokumentet ändras eller presenteras i sådan form eller i sådant sammanhang som är kränkande för upphovsmannens litterära eller konstnärliga anseende eller egenart.

För ytterligare information om Linköping University Electronic Press se förlagets hemsida `http://www.ep.liu.se/.`

Copyright

The publishers will keep this document online on the Internet – or its possible replacement – for a period of 25 years starting from the date of publication barring exceptional circumstances.

The online availability of the document implies permanent permission for anyone to read, to download, or to print out single copies for his/her own use and to use it unchanged for non-commercial research and educational purposes. Subsequent transfers of copyright cannot revoke this permission. All other uses of the document are conditional upon the consent of the copyright owner. The publisher has taken technical and administrative measures to assure authenticity, security and accessibility.

According to intellectual property law, the author has the right to be mentioned when his/her work is accessed as described above and to be protected against infringement.

For additional information about the Linköping University Electronic Press and its procedures for publication and for assurance of document integrity, please refer to its www home page: `http://www.ep.liu.se/.`

© Jonas Hautamäki Bengtsson
Samuel Nikanorsson Sundmark

Abstract

Emergency departments operate under high pressure and information load, placing significant demands on clinical leadership to maintain situation awareness. At the emergency care unit of Linköping University Hospital, patient information is presented through a spreadsheet-like interface that, while informationally complete, requires substantial manual aggregation and reliance on individual experience to form an overview. This thesis investigates how alternative visualization methods and interface design principles can improve the presentation of patient information in this setting.

Following a human-centered design approach structured around the Design Thinking framework, semi-structured interviews with charge nurses and lead physicians were analyzed thematically to identify the primary usability challenges of the current interface. These challenges informed the design of a high-fidelity dashboard prototype, which presents current patient information against historical baselines and whose five core elements were grounded in established theory on cognitive load, situation awareness, and visual scanning behavior. The prototype was evaluated with clinical staff after a short demonstration, using a survey based on the Technology Acceptance Model, complemented by qualitative feedback.

The analysis identified four recurring challenges: high extraneous cognitive load, a lack of historical context, information fragmentation, and limited support for forward-looking assessment. The dashboard was received positively, with high scores for perceived ease of use and moderately high scores for perceived usefulness, while the factors most likely to affect broader adoption concerned the validity and transparency of the underlying data rather than the interface itself. Given the exploratory scope, the small sample, and the short evaluation horizon, these findings should be read as indicative. The principal contribution of the thesis is a grounded foundation – a set of identified needs, a theoretically motivated design rationale, and an initial indication that the approach merits further development.

Acknowledgements

We thank our supervisor Jens Wretborn at Linköping University Hospital for the opportunity to conduct this Bachelor's thesis, and our examiner Emma Enocksson Svensson together with our second supervisor Klas Arvidsson at the Department for Computer and Information Science at Linköping University for their guidance throughout the project. We are grateful for the physicians and nurses at Linköping University Hospital's emergency department for the time and insights they shared with us, and to Ludwig Halvorsen for valuable advice on user experience and user interface design.

Acknowledgements **3**
Abbreviations **7**
1. Introduction **8**
1.1 Background 8
1.2 Purpose 8
1.3 Research Questions (RQs) 8
2. Theory **9**
2.1 Human-Centered Design and Design Thinking 9
2.2 Situation Awareness 10
2.3 Cognitive Load & Information Visualization 11
2.3.1 Cognitive Load Theory 11
2.3.2 Principles of Information Visualization 12
2.3.3 Visual Scanning Behavior 12
F-pattern 12
Z-pattern 13
2.3.4 Baselines 14
2.4 Thematic Analysis 14
2.5 Technology Acceptance Model 15
2.5.1 Perceived Usefulness 15
2.5.2 Perceived Ease of Use 15
2.5.3 Relationship Between the Constructs 15
3. Method **17**
3.1 Data Collection 17
3.1.1 Selection of Participants 17
3.1.2 Semi-structured Interviews 17
3.2 Data Analysis 19
3.2.1 Transcription and Data Extraction 19
3.2.2 Thematic Analysis 19
3.3 Idea Generation and Prioritization 20
3.3.1 Brainstorming & Element Selection 20
3.3.2 Expert Consultation and Validation 20
3.4 Prototyping 20
3.5 Evaluation 22
3.5.1 User Testing 22
3.5.2 Question Design 23
3.6 Quality and Ethical Considerations 23
4. Results **24**
4.1 Identified Usability Challenges 24
4.1.1 Cognitive Overhead in the Current Interface 24

4.1.2 Lack of Historical Context 24
4.1.3 Fragmented Information and Workflow Disruption 25
4.1.4 Absence of Forward-Looking Information 25
4.1.5 Summary of Identified Challenges 25
4.2 Dashboard Design 26
4.2.1 Information Architecture and Layout 26
4.2.2 Element 1: Patient Load 27
4.2.3 Element 2: Patients by Age Span 28
4.2.4 Element 3: Patients by Priority (Triage) 29
4.2.5 Element 4: Patient Flow Trends 30
4.2.6 Element 5: Team-Level Aggregation 30
4.2.7 Summary of Design Rationale 31
4.3 Evaluation Through the Technology Acceptance Model 32
4.3.1 Overall TAM Scores 32
4.3.2 Item-level Results 33
4.3.3 Specific Visualization Items 34
4.3.4 Qualitative Findings on Adoption Factors 34
4.3.5 Summary of TAM Evaluation 35
5. Discussion 36
5.1 Interpretation of Findings 36
5.1.1 RQ1: Usability Challenges in the Current Interface 36
5.1.2 RQ2: Visualization of Key Patient Variables 36
5.1.3 RQ3: Perceived Acceptance and Factors Influencing Adoption 37
5.2 Theoretical Implications 38
5.3 Design Reflections 39
5.3.1 Information Density 39
5.3.2 The Team-Level Indicator as a Minimal Overview Solution 39
5.3.3 Ordering of the Triage Bars 40
5.3.4 Placement of Triage Relative to Scanning Patterns 40
5.3.5 System Familiarity 40
5.3.6 Scope in Relation to the Wider Information Environment 40
5.4 Methodological Reflections and Limitations 41
5.4.1 Sample and Evaluation Context 41
5.4.2 Data Collection and Analysis 41
5.4.3 Research Approach and Tooling 42
5.4.4 Framing of the Project 43
5.5 Sustainability and Ethical Considerations 44
5.5.1 Sustainability 44
5.5.2 Ethical Considerations 44
5.6 Source Criticism 45
5.7 Future Work 46

5.7.1 Baseline Computation..... 46
5.7.2 Empirical Analysis of Historical Data..... 46
5.7.3 Additional Clinical Variables..... 46
5.7.4 Cross-Sectional Overview Across Teams..... 47
5.7.5 Automated Flagging and Decision Support..... 47
5.7.6 Longitudinal Evaluation..... 47
6. Conclusion..... 48
References..... 49
Appendix..... 51
[A] Prototyping prompts..... 51

Abbreviations

AI	Artificial Intelligence
CN	Charge Nurse
HCD	Human-Centered Design
LP	Lead Physician
PEOU	Perceived Ease of Use
PU	Perceived Usefulness
RQ	Research Question
SA	Situation Awareness
TA	Thematic Analysis
TAM	Technology Acceptance Model

1. Introduction

1.1 Background

Emergency departments operate in environments where large volumes of patients and information must be managed simultaneously, placing high demands on healthcare personnel to maintain Situation Awareness (SA) and make timely decisions. In the observed clinic, Linköping University Hospital emergency care unit, approximately 100–200 patients are handled daily, with 10–60 patients actively present in the system at any given time. To support this workflow, patient information is currently presented through the patient overview in the COSMIC Electronic Health Record (EHR) system (Cambio Healthcare Systems), an interface that largely resembles a spreadsheet (recreated in Figure 1). Each row represents a patient and contains information supporting prioritization – such as triage urgency color, age, presenting complaint, and progress within the care process – along with administrative details such as assigned team, location, arrival mode, and responsible staff.

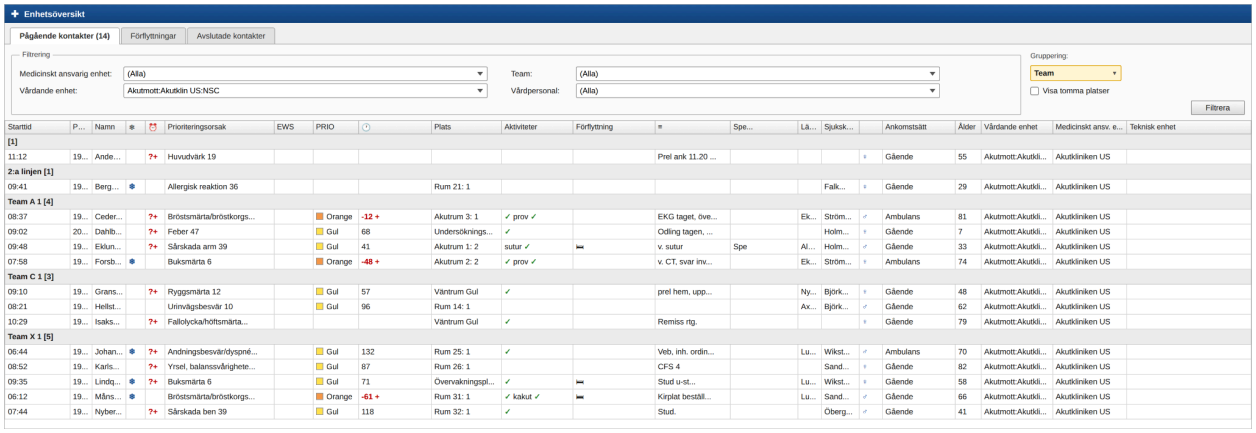


Figure 1. Recreation of the current patient overview (Enhetsöversikt) in the COSMIC EHR at the Linköping University Hospital emergency care unit. The layout and column structure reflect the actual interface; all patient, staff and case data are fictitious.

Although the necessary data is available in the system, challenges arise in how the information is presented and used in daily practice. Healthcare staff must continuously monitor patient status, identify delays, and detect potential omissions in the care process while simultaneously handling a high workload in a time-critical environment. Under these conditions, interfaces that require extensive manual scanning and interpretation may limit the ability to quickly obtain an overview of the current situation.

The current solution provides access to relevant information, but acquiring an overview of the emergency department's state comes with considerable overhead – mental arithmetic, filtering out the information that matters, and reliance on verbal coordination with colleagues – and it leans heavily on individual experience. Important details may therefore be overlooked, and delays in patient flow may become difficult to identify in time. The problem is not primarily related to data availability, but rather to how information is structured, visualized, and prioritized within the interface.

1.2 Purpose

This thesis therefore investigates how alternative visualization methods and interface design principles can improve the presentation of patient information in Linköping University Hospital's emergency care unit. The work focuses on developing interface concepts that support clearer overview, improved prioritization, and better usability in a complex and dynamic clinical environment.

1.3 Research Questions (RQs)

RQ1: What are the primary usability challenges for Charge Nurses (CNs) and Lead Physicians (LPs) when forming an overview of the department's state using the current spreadsheet-like interface?

RQ2: How can key patient variables – such as priority levels and age – be visualized to reduce cognitive load and enhance situation awareness for department leaders?

RQ3: To what extent do clinical leaders perceive the proposed dashboard as useful and easy to use, and what factors influence its potential adoption in the emergency department?

2. Theory

This chapter will lay the foundation of this study's research design, analysis approach and the development of the system. The theories presented in this chapter are not independent from each other but together form a coherent groundwork for what our method is based on. The leadership role in the emergency care system demands a continuous overview with high-quality SA under high cognitive pressures. These theories will address why it is difficult without decision support, how information can be designed to alleviate that difficulty, how to structure the development process to focus on the users' needs, how to interpret qualitative data from users' interviews and how the resulting system's likelihood of adoption was assessed.

2.1 Human-Centered Design and Design Thinking

Human-Centered Design (HCD) is an approach to systems design and development [1]. A core principle of HCD is that design decisions are based on an explicit understanding of users, their tasks, and the environments in which the system is used, rather than on assumptions [1].

Design Thinking [2] is an iterative, human-centered problem-solving methodology. This thesis follows the five-stage model proposed by the Hasso Plattner Institute of Design at Stanford University (the d.school). It is a structured yet flexible approach in which the end user's needs and experience shape the development process. The framework consists of five stages: Empathize, Define, Ideate, Prototype, and Test; while presented as sequential, they are intended to be applied in a non-linear and iterative manner, as shown in figure 2.

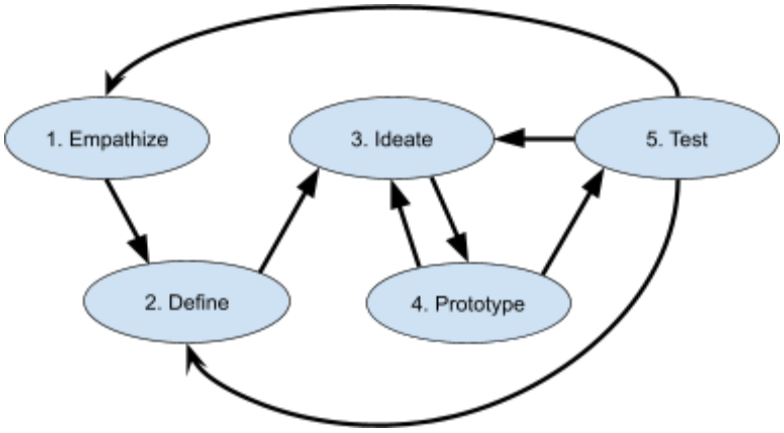


Figure 2. The Non-Linear Design Thinking Process visualized. The arrows illustrate the Non-Linear directions the different steps can take. Own illustration, adapted from Design thinking [2].

Stage 1: Empathize

The first stage involves getting to know the users, their behaviors, and their pain points in order to gain a deep understanding of what is being developed and why. Through observation and interviews, the researchers identify how users interpret the existing system, where potential bottlenecks or problems lie, and what is important for the system to work. This stage establishes a foundation grounded in understanding rather than assumptions.

Stage 2: Define

The second stage synthesizes the insights gathered during the Empathize stage into clear, human-centered problem statements. It narrows the scope of the project and aligns development around a shared objective.

Stage 3: Ideate

With a clearly defined problem in hand, the third stage encourages the generation of a wide range of ideas and solutions without initial constraints. This draws on creative problem-solving methods such as brainstorming, which relies on divergent thinking – generating ideas freely before any are judged. When working in pairs or teams, each member first explores solutions independently before converging to compare and discuss ideas [3]. The purpose of this stage is to explore the solution space as broadly as possible while deferring judgement, rather than to evaluate the feasibility of the ideas.

Stage 4: Prototype

In the fourth stage, selected ideas are translated into testable representations, such as mock-ups or interactive prototypes, that can be evaluated without heavy development investment. Testing these prototypes surfaces potential issues early, before substantial development effort has been committed.

Stage 5: Test

In the fifth stage, prototypes are tested with end users, and their feedback guides refinements, reveals new problems, and indicates how usable the design is. The iterative loop between prototyping and testing is an effective way to converge on the users' needs and the system requirements.

2.2 Situation Awareness

Endsley defines situation awareness as '*the perception of the elements in the environment within a volume of time and space, the comprehension of their meaning, and the projection of their status in the near future*' [4]. This framework divides SA into three levels – perception, comprehension, and projection – and applies to any environment that requires real-time decision-making under dynamic conditions.

Level 1 - Perception
At the first and most fundamental level, the individual perceives the relevant elements in their environment. In the context of emergency care, this corresponds to an LP or CN observing, for example, that fourteen patients are currently registered in the department.

Level 2 - Comprehension
At the second level, the individual interprets and integrates those elements to understand what they mean in relation to their goals. Building on the previous level, fourteen patients is in itself neither good nor bad; to interpret the number, the individual must understand its significance relative to the current context and their goals. Without this comprehension, the individual cannot move beyond perception to assess what the situation means.

Level 3 - Projection
At the third level, the individual uses that understanding to anticipate how the situation will develop. For example, a CN who recognizes that the current patient volume is 50 percent above the norm for that time of day can project that the department is unlikely to keep up with the inflow unless action is taken. Such a projection allows problems to be addressed proactively rather than reactively.

2.3 Cognitive Load & Information Visualization

Cognitive Load Theory and information visualization complement each other in informing how information should be structured and presented in high-pressure environments. Cognitive Load Theory identifies three types of load on the limited capacity of human working memory – intrinsic, extraneous, and germane – and shows how poorly designed interfaces increase the risk of errors and weaken overview. Information visualization, in turn, describes how people scan and interpret what they see. Together, these theories ground the decisions of how much to show and how to present it comprehensibly.

2.3.1 Cognitive Load Theory

Cognitive Load Theory, originally developed by Sweller [5], holds that human working memory is limited in capacity for processing new information. In a later extension, Sweller, van Merriënboer, and Paas [6] distinguished between three types of cognitive load: intrinsic, extraneous, and germane.

- **Intrinsic load** - effort that comes from the inherent complexity of the task itself and cannot be reduced without simplifying the task.

- **Extraneous load** - effort caused by how information is presented rather than the task itself; for instance, a poorly organized display forces the reader to do work that a well-organized one would spare.
- **Germane load** - effort devoted to genuinely processing the material and building the mental schemas that support understanding; the productive mental work that learning and sound judgement depend on.

When a task exceeds what working memory can process at once, performance degrades: decisions become slower, errors more frequent, and important information more likely to be missed. In emergency departments, where LPs and CNs must simultaneously monitor patient flows, respond to incoming calls, and manage staff capacity, the risk of cognitive overload is not incidental but an inherent consequence of the role and the high-pressure environment in which it is performed.

Effective interface design aims to minimize extraneous load so that working memory can be devoted to intrinsic and germane processing instead. This framework motivates the design approach taken in this study. The current spreadsheet-like interface places much of this work on the user: aggregating scattered values into an overall picture, and drawing on experience and memory to judge whether a given situation is normal. Both are extraneous loads – mental work imposed by how the information is presented rather than by the task itself. A dashboard that aggregates this information and presents it visually against historical values can carry that work in the system instead, freeing working memory for the intrinsic and germane processing the task actually requires.

2.3.2 Principles of Information Visualization

Research on information visualization offers established principles for how data can be presented to reduce extraneous cognitive load and support faster, more accurate comprehension. Several of these are particularly relevant to designing a dashboard for an emergency department.

Pre-attentive attributes are visual properties such as color, shape, and size that the human visual system processes subconsciously [7]. Encoding critical information with these attributes – for example, using color to show that a metric is falling or that a value lies outside its normal historical range – lets the user detect anomalies before any deliberate analysis. This is valuable in time-critical environments such as the emergency department, where sustained analytical attention is not always available.

Color-coding is a well-established pre-attentive technique commonly used in healthcare settings [8]. When a color is applied consistently and carries a clear meaning, for example, triage priority levels, it lets staff recognize a patient's priority more quickly.

Tufte's data-ink principle [9] holds that as much as possible of a display's visual elements should represent actual data, while non-data elements such as decoration and unnecessary gridlines should be minimized. Such elements add visual noise and thereby increase extraneous cognitive load.

2.3.3 Visual Scanning Behavior

The placement of elements within an interface has a measurable effect not only on how quickly the user can achieve SA but also on what information they attend to first. Two established models of visual scanning behavior are relevant to this study:

F-pattern

The F-pattern [10] describes how users read text-heavy interfaces in three sequential movements, as shown in figure 3. First, the eyes scan horizontally across the top of the display (1); then they make a shorter horizontal sweep slightly below (2); and finally they scan vertically down the left-hand side (3). The top of the display, and the upper-left in particular, therefore receives the most attention, making the top row the most valuable area for the highest-priority information.

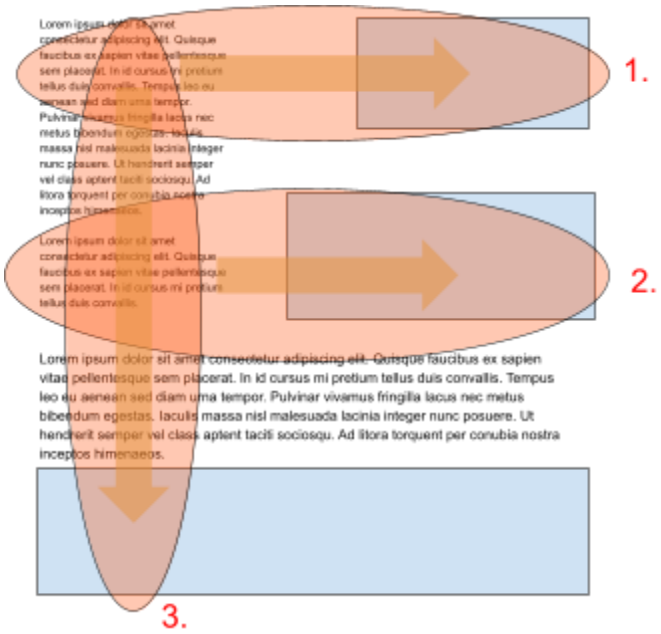


*Figure 3. A visual representation of the **F-pattern**; the numbers represent the order the users perceive step by step.*

Z-pattern

The **Z-pattern** [11] describes the scanning behavior in more visually structured layouts, as shown in figure 4. The eye first travels from the top-left to the top-right (1), then diagonally down to the bottom-left (2), and finally across to the bottom-right (3).

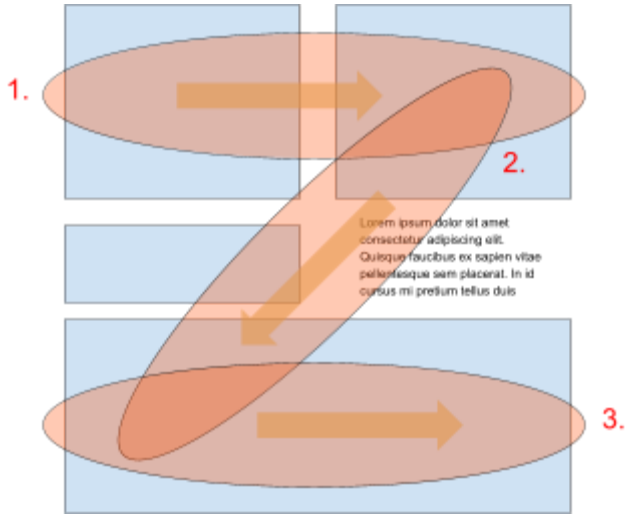


*Figure 4. A visual representation of the **Z-pattern**; the numbers represent the order the users perceive step by step.*

Although the F-pattern and the Z-pattern apply to different types of layout, both begin along the top of the display, moving from the upper-left to the upper-right before descending. The top row is therefore the first region a user scans regardless of which pattern a layout follows, which makes it the most reliable position for the information that matters most.

2.3.4 Baselines

Endsley's model establishes that comprehension (Level 2) requires interpreting perceived data in relation to one's goals and context [4], but it does not prescribe how this reference point should be formed in a given domain. For patient flow, we adopt the notion of a baseline: an expected value against which current conditions can be evaluated.

Such a baseline must account for the fact that patient flow is not distributed uniformly over time: a patient count that is normal at midday on a Monday may be unusually high at midday on a Saturday. A meaningful baseline therefore reflects the expected load for a comparable time and context, rather than a single average across all periods.

2.4 Thematic Analysis

Thematic Analysis (TA) is a qualitative method for identifying, analysing, and interpreting patterns of meaning, or themes, within a dataset [12]. While the approach predates their work, Braun and Clarke systematised and operationalised it into a clearly defined procedure, and it is now widely adopted across research domains.

Unlike methods tied to a specific theoretical framework, TA is flexible: it can be applied from a range of theoretical positions and adapted to suit the nature of the research question [12]. This makes it well suited to exploratory studies, where the goal is to surface and understand the user's environment, behaviours, and perspectives.

The method follows six phases:

1. Familiarisation with the data.
2. Generation of initial codes.
3. Searching for themes.
4. Reviewing themes.
5. Defining and naming themes.
6. Producing the report.

Themes are not identified by frequency alone but assessed by their relevance and significance to the research question [12].

In this project, TA provides a structured means of interpreting the qualitative data gathered during the Empathize stage of the Design Thinking Process. By identifying recurring themes, TA translates raw user insights into clearly defined problem areas that feed directly into the Define stage.

2.5 Technology Acceptance Model

Predicting whether users will accept and adopt a new information system is a long-standing concern in the information systems field. Early approaches drew on expectancy theory – the idea that people are motivated to act when they expect the action to produce a desired outcome [13] – but the measures used to operationalize it correlated inconsistently with actual system use, a problem Davis [14] attributed to a lack of validated instruments rather than to the theory itself. Davis [14] addressed this by developing and validating two constructs, Perceived Usefulness (PU) and Perceived Ease Of Use (PEOU), proposed as the fundamental determinants of user acceptance and forming the foundation of the Technology Acceptance Model (TAM) [15].

2.5.1 Perceived Usefulness

Davis [14] defines perceived usefulness as “the degree to which a person believes that using a particular system would enhance his or her job performance”. The construct captures the user’s subjective judgement of the system’s instrumental value – whether it makes work faster, improves the quality of output, or increases productivity. In an organizational context, this is theorized to be the primary motivation for adoption, since performance gains are linked to tangible rewards such as recognition and professional advancement.

2.5.2 Perceived Ease of Use

Davis [14] defines perceived ease of use as “the degree to which a person believes that using a particular system would be free of effort.” This construct reflects the cognitive and physical demands of interacting with the system. Davis links it to Bandura’s self-efficacy theory [16] – individuals’ judgements of how well they can carry out a course of action – arguing that ease of use draws on the same underlying belief

in one's own capability [14]. A system perceived as requiring substantial mental effort to operate therefore creates a barrier to adoption independent of its functional value.

2.5.3 Relationship Between the Constructs

The central finding of Davis [14] is that PU and PEOU are two distinct concepts that contribute differently to user intention. PU was more strongly correlated with self-reported system use, that is, users' own descriptions of their usage behavior, than PEOU was, across both studies conducted. When usefulness was statistically controlled for, ease of use largely ceased to predict usage on its own. This suggests that ease of use influences usage indirectly; by making the system easier to use, users are also more likely to perceive it as useful, and it is that perceived usefulness that ultimately drives adoption.



Figure 5. Hierarchy of the TAM model simplified [15].

The hierarchy shown in figure 5 has importance in design evaluation. Systems that score high in ease of use but low on usefulness are unlikely to be implemented or adopted, because users are motivated by what the system does for them rather than how easy it is to operate. As Davis noted, users are often willing to tolerate some degree of difficulty if a system provides critically needed functionality, whereas no level of ease of use can compensate for a system that does not perform a useful function.

Davis developed and validated measurement scales for both constructs through a multi-stage process involving content validity pretesting, a field study, and a laboratory study. The final scales consist of six items per construct, measured on a seven-point semantic differential scale with the anchors *likely* and *unlikely,* see figure 6. Across these validation studies the scales showed high internal reliability and strong validity [14].

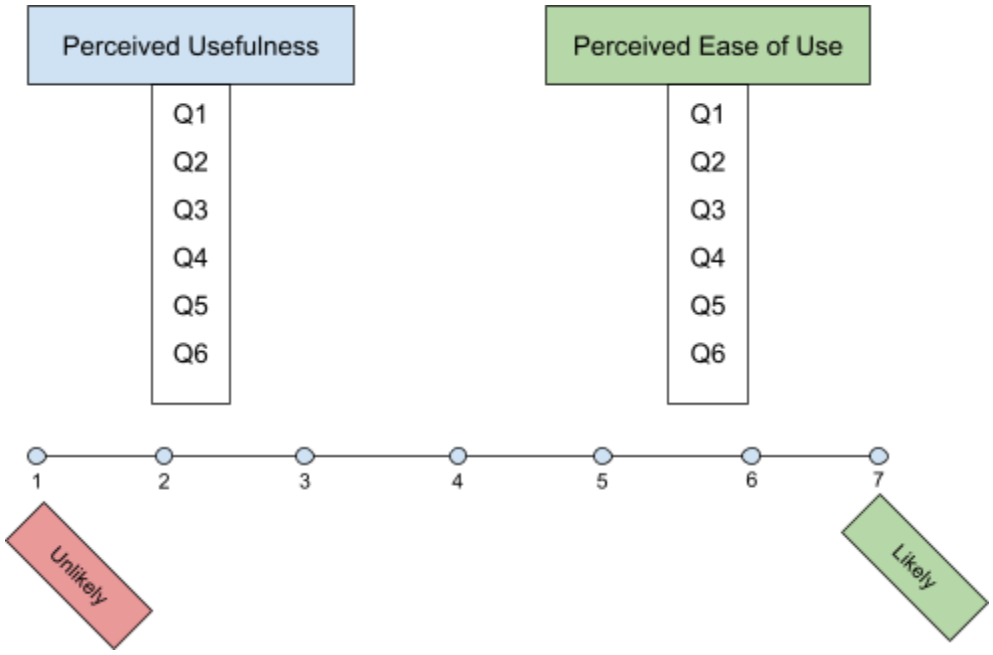


Figure 6. A visualization of the TAM scale structure, where Q1–Q6 represent the six items per construct and the 1–7 scale represents the respondent's rated likelihood of each statement being true.

TAM is well suited to this study on two counts. Substantively, the project examines a digital support interface for an emergency department, where adoption depends on whether clinical staff find the system both useful and easy to use in everyday work. Here, PU concerns whether the interface supports rapid SA, prioritization, and patient-flow management, while PEOU concerns whether that information can be accessed without unnecessary cognitive effort under time pressure.

Methodologically, the final stage of the project uses a survey based on TAM constructs, which lets the interview findings and the resulting design be evaluated through established measures of PU and PEOU. This connects the qualitative account of user needs with a quantitative assessment of how well the design is accepted.

3. Method

This study employs a qualitative and iterative research design grounded in HCD principles [1]. To address the complex cognitive demands of leadership roles within emergency care, the Design Thinking framework was utilized as the methodological backbone [2]. This approach was chosen to ensure a deep understanding of the end-users' current workflows and mental models before initiating the technical design phase.

The project began with a scoping meeting that established and anchored the assignment. The meeting brought together the authors, the examiner and supervisor at the Department of Computer and Information Science at Linköping University, and the supervisor at the emergency department of Linköping University Hospital, who had originally proposed the project. The authors presented their understanding of the task, and the scope was refined into a shared definition of the work and confirmed to meet the requirements of the thesis. The agreed aim was to investigate the contextual and cognitive challenges faced by LPs and CNs in forming an overview through the current spreadsheet-like interface, and, on that basis, to design a supplementary dashboard combining real-time data with historical baselines in order to reduce administrative overhead and cognitive load.

As an initial step in building domain understanding, the supervisor at Linköping University Hospital – who divides his time between working as LP and performing research – gave the authors a contextual walkthrough of the current system, drawing on his own experience in that role. He demonstrated how information such as triage priority, patient age, and chief complaint is manually aggregated and interpreted to assess the department's overall workload. This orientation helped the authors identify the data points most relevant to high-stakes decision-making and informed the subsequent data collection and design requirements.

3.1 Data Collection

The data collection phase was structured to gather deep insights into both the functional requirements of the dashboard and the contextual environment of the emergency department.

3.1.1 Selection of Participants

Participants were identified and recruited in collaboration with the supervisor at Linköping University Hospital. A purposeful sampling approach [17] was used to ensure that all participants held the specific roles of either LP or CN. These individuals were selected for their extensive experience with the current overview interface and their direct responsibility for managing patient flow. Seven participants were recruited based on their availability during the researchers' site visits, and all seven held the relevant leadership roles of LP or CN. One of the seven was the project's supervisor at Linköping University Hospital, who also took part in his capacity as an LP.

3.1.2 Semi-structured Interviews

The interviews were conducted during the first week of the study. Because the study was exploratory and the authors did not yet have a precise picture of what they were looking for, high-quality data was

prioritized to minimize the risk of missing critical information. The researchers therefore adopted specific roles during the sessions: one researcher acted as the primary interviewer, while the other served as a scribe, taking detailed notes.

The primary objective was to collect comprehensive data regarding the functionality of the current interface, the user experience of LPs and CNs, and the general environmental factors influencing the emergency department's workflow.

Ethical Considerations and Introduction

Each session began with a standardized introduction to establish a professional and safe environment. Participants were informed about the study's purpose, their right to voluntary participation, and guaranteed total anonymity in the final report. Informed consent was obtained for audio recording, with the assurance that all recordings would be destroyed following transcription.

Thematic structure and Probing

The interview guide was divided into two main thematic areas:

- **Theme A: Situation Awareness and Workflow:** Questions focused on how participants build an initial "mental image" of the department's status upon starting a shift and which data points are critical for this assessment.
- **Theme B: Identifying Issues and Future Needs.** This section aimed to identify critical bottlenecks and pain points in the current workflow. Drawing on the Critical Incident Technique [20], participants were asked to describe specific situations where gaining an overview was difficult and to envision real-time decision support features.

To deepen the insights, the researchers drew on several of the question types described by Kvale and Brinkmann [19]:

- Specifying questions to elicit concrete examples.
- Deliberate silence to encourage further elaboration.
- Interpreting questions to verify the researchers' understanding of participant statements.

Iterative Refinement of the Interview Process

A key aspect of the data collection was the iterative refinement of the interview guide and technique based on early experiences. The first two participants (P1 and P2) were purposefully selected by the clinical lead to represent diverse personalities and cognitive styles.

Following the first interview, it was observed that the participant was overly analytical, leading to hesitation. To counter this, a specific prompt was added to subsequent interviews, encouraging a "stream of consciousness" approach where participants were asked to speak freely without overthinking. After the second interview, which was more conversational and expansive, the interviewers realized the value of allowing participants to "drift" into related topics, as this provided rich contextual data that was later captured through transcription.

Consequently, from P3 through P7, the interview style shifted from a rigid Q&A format toward a more conversational inquiry [19]. This allowed for a more relaxed atmosphere, facilitating deeper insights into the participants' actual needs and mental models. The overarching goal of this data-heavy approach was to gather raw material for the subsequent extraction of discrete, actionable insights used in the analysis phase.

3.2 Data Analysis

The analysis phase was designed to systematically transform raw qualitative data into actionable design requirements. This was achieved through a multi-stage process involving automated transcription, independent coding, and collaborative synthesis.

3.2.1 Transcription and Data Extraction

To prepare the data for analysis, the audio recordings were transcribed using OpenAI's Whisper [22], specifically the large-v3 model released in November 2023 [23] via the software Subtitle Edit. This automated process provided a high-fidelity textual foundation, which was subsequently refined through manual review.

The extraction of initial insights was conducted independently by each researcher. By reviewing the transcripts and audio recordings separately before comparing interpretations, the researchers aimed to reduce the influence of any single researcher's bias and to avoid premature consensus – an approach consistent with analyst triangulation [17]. Following the principles of inductive TA [12], the goal during this stage was not to categorize the data according to pre-defined themes, but to capture as many discrete observations, pain points, and requirements as possible. This exhaustive extraction resulted in a total of 144 initial data points, including duplicates and peripheral observations.

3.2.2 Thematic Analysis

The subsequent analysis followed an iterative process inspired by Affinity Diagramming [21]. Each researcher transferred their independent findings onto physical notes, which were organized into two main thematic pillars with corresponding sub-themes:

- **Work Environment:** Focusing on Pain points, Information flow, and Organizational structure.
- **System Interaction:** Focusing on Current pain points, Available data, Contextual background, and Design ideas/proposals.

After this initial individual categorization, the researchers converged to synthesize their findings. This collaborative phase involved:

1. **Comparison and Discussion:** Reviewing each other's themes and individual insights to identify overlaps and unique observations.
2. **Deduplication and Filtering:** Removing redundant entries and filtering out irrelevant data (the "trash" pile), while retaining the original context for future reference.

This process resulted in a final set of 108 unique insights.

3.3 Idea Generation and Prioritization

To transition from analysis to design, a phase of divergent and convergent thinking was initiated [3]. The researchers began by revisiting the established themes to ensure the empirical data remained central to the creative process.

3.3.1 Brainstorming & Element Selection

An initial brainstorming session was conducted where the researchers worked independently to generate as many design solutions as possible. This independent approach was chosen to foster diverse perspectives, resulting in one set of ideas focused on data-driven logic and another on visual components and layout.

The resulting ideas were compared and synthesized. Components were decomposed into their smallest functional units and evaluated through a point-based prioritization process. Each element was ranked on a scale of 1 to 5 based on its perceived necessity (where 1 represented “low relevance” and 5 represented “essential”). This process narrowed the 108 initial unique data points down to 12 high-priority features, which were further distilled into five core dashboard elements.

To optimize the information architecture of the dashboard, the layout was informed by established visual scanning behaviors: The F-pattern and the Z-pattern [10, 11]. The F-pattern describes how users interact with text-heavy interfaces by scanning horizontally across the top and then vertically down the left side. Conversely, the Z-pattern is typical for visually-oriented layouts, where the eye moves from the top-left to the top-right, then diagonally down to the bottom-left and across to the bottom-right. By applying these patterns, the five core elements were positioned to ensure that the most critical real-time data is captured within the initial scanning path of the LPs and CNs, thereby reducing the time required to achieve SA.

3.3.2 Expert Consultation and Validation

Before proceeding to high-fidelity prototyping, the proposed core elements were validated through expert consultation. The researchers consulted a user experience and user interface design coach for guidance on interface best practices and the project supervisor (the clinical lead) to ensure the selected elements aligned with the clinical reality of the emergency department. This step ensured that the design requirements were both technically sound and contextually relevant.

3.4 Prototyping

The development of the prototype followed an iterative high-fidelity approach using Lovable, an Artificial Intelligence (AI)-driven prototyping platform that generates and revises a running web application from natural-language instructions. Lovable was selected because it allowed a responsive, interactive dashboard to be produced and refined rapidly without committing substantial development effort, in keeping with the role of prototyping in the Design Thinking Process (Section 2.1). The platform generated a React application styled with Tailwind CSS and using shadcn/ui components and lucide-react

icons; the synthetic data was held as hardcoded values within the application rather than served from a backend, since the prototype was intended for demonstration rather than deployment. Because the platform is generative and non-deterministic, the same prompts are not guaranteed to reproduce an identical result; the process is therefore documented here at the level of the instructions given, with the full prompt sequence reproduced in Appendix [A] to support replicability as far as the tool permits.

The authors' direct input to the build specification was a set of core elements identified in Section 3.3.1 – what the dashboard should contain – which followed from the analysis and prioritization in sections 3.2-3.3. These elements were then expanded into a single structured build prompt with the assistance of an AI language model, which supplied much of the concrete presentation detail: the fixed, non-scrolling layout, the arrangement of the elements, the pre-bar dashed baselines, the two-series trend charts with the historical curve extended beyond the current time, the timeframe selector, and the styling conventions. The authors reviewed and accepted this specification before it was issued to the platform. The theoretical framework set out in Chapter 2 was then applied to assess and justify these choices – for example the positioning of the most critical metric relative to the F- and Z-pattern scanning paths (section 4.2.1) and the reservation of color for triage priority (section 4.2.3) – consistent with the abductive logic described in Section 5.4.3, in which design decisions and theory were brought into alignment iteratively rather than derived in a fixed sequence.

A second prompt shaped the synthetic data so that the distribution across the four teams reflected plausible clinical allocation patterns rather than uniform values – for example an older, higher-acuity cohort for one team and a younger, lower-acuity cohort for another – and required these patterns to be reflected in the individual synthetic patient records, not only in the aggregate charts. This produced a more realistic basis for the demonstration while remaining entirely synthetic and non-identifiable, consistent with Section 3.6.

The remaining prompts were targeted refinements of the generated interface, addressing among other things the legibility of the load figures, the color-coding of the inflow and outflow totals to match their respective trend lines, and the placement of the timeframe selector. One refinement is illustrative of the iterative loop: an initial attempt to make the patient-load deviation salient through a shaded band above and below the average line was judged to add visual noise and was replaced, in the following prompt, with a compact colored badge behind the numeric deviation label – red when the current load is above the historical average and green when below – leaving the chart area uncluttered, in line with the data-ink rationale of section 2.3.2.

A final change concerned team navigation. The original prototype interface presented team selection through a dropdown; this was subsequently replaced, through two further prompts, with an explicit row of buttons for the overall view and each individual team. As described in section 5.4.2, this followed an observation made during the first evaluation sessions, where participants struggled to return to the department-wide view, and the platform allowed the change to be made between sessions.

3.5 Evaluation

To assess the perceived usefulness, perceived ease of use , and potential adoption of the proposed dashboard, an evaluation was conducted based on TAM [15], complemented by qualitative feedback. TAM provides the two fundamental constructs PU and PEOU [14] for capturing a user's assessment of a system after exposure to it.

The evaluation was carried out in person at the Linköping University Hospital emergency department, where the participants work, in six individual sessions with one participant per session. Each session lasted approximately 15 to 20 minutes. The participants were largely different individuals from those interviewed during the Empathize phase; one participant – the clinical academic supervisor – took part in both phases. The group was not restricted to staff holding a formal LP or CN role: it included clinical staff working within teams as well as a physician manager, all recruited on the basis of having a working relationship with the current interface.

Each session began with a short introduction in which the researchers presented themselves and described the work. The purpose of the session was deliberately framed to orient participants toward assessing whether the type of data and its visualization could provide value in their work, rather than toward evaluating the prototype as a finished product. This framing was a methodological choice intended to direct attention to the underlying concept rather than to incidental shortcomings of an early prototype. The session then proceeded to a walkthrough of the dashboard, after which the participant completed a survey based on TAM constructs and adapted to the context of emergency care decision support.

3.5.1 User Testing

After the introductory framing, the researchers walked the participant through all elements of the dashboard, including the five core elements and the use of baselines. The degree of direct interaction varied between sessions: in some sessions a researcher navigated the prototype while demonstrating it, while in others the participant navigated it themselves. Throughout the walkthrough participants were free to comment and ask questions, and the sessions were audio-recorded and subsequently transcribed. The free-text responses in the survey therefore partly overlap with comments made aloud during the walkthrough.

When the walkthrough was complete, participants were asked whether they felt sufficiently familiar with the system to answer the survey; all confirmed that they did. Each participant then completed the survey independently on a computer while the researchers waited without intervening, with the option to ask questions if anything was unclear. Once the survey was submitted, the session concluded.

3.5.2 Question Design

The survey instrument was structured into seven parts. Part 1 collected background information. Parts 2 through 6 consisted of statements rated on a seven-point scale ranging from “strongly disagree” to “strongly agree”, and part 7 consisted of open-ended free-text questions.

The five scaled parts comprised 22 items in total: six items for PU (part 2), six for PEOU (part 3), four for Attitude Toward Use (Part 4), three for intention to use (Part 5), and three items addressing specific visualization choices in the dashboard (Part 6). The four free-text questions in Part 7 addressed what worked best in the prototype, what participants would change or improve, what information they felt was missing, and whether they foresaw obstacles to introducing the tool in daily practice. The analysis reported in Chapter 4 focuses on the PU and PEOU constructs together with the three visualization-specific items, complemented by the qualitative feedback on adoption factors. The Attitude and Intention items were collected as part of the same instrument but are not analyzed further here.

The standard TAM items were adapted to the clinical setting. For instance, the usefulness construct included the item “Using the dashboard in my work would give me a faster overview of the situation in the department”. In a deviation from the original instrument validated by Davis [14], which uses a seven-point semantic differential scale with the anchors *likely* and *unlikely*, the present survey used a seven-point agreement scale. This adaptation kept the response format consistent across all five scaled parts and aligned with the way the statements were phrased, but it means the instrument is an adaptation of the TAM scales rather than a replication of the original measurement format. For brevity, this evaluation is hereafter referred to simply as the TAM evaluation.

Two limitations of the Part 1 background items should be noted, as they constrain the role-based interpretation discussed in section 5.1.3. First, the item capturing professional role offered response options that mixed leadership function with professional profession, so the responses do not reliably indicate which participants held a formal LP or CN role at the time of evaluation. Second, the item on experience asked for years of experience in the LP or CN role rather than years of experience working with the current interface; the latter would have been more directly relevant to the system the dashboard is intended to supplement.

3.6 Quality and Ethical Considerations

Research Quality and Rigor

The quality of the qualitative analysis was upheld through analyst triangulation; researchers conducted the initial coding of transcripts independently to reduce individual bias before synthesizing the final themes. In terms of sample size, the recruitment of seven participants sits just above the point at which Guest et al. found that a majority of codes (73%) had emerged in relatively homogeneous samples – the first six interviews [18]. While this does not constitute full thematic saturation, which in their study was approached only around twelve interviews, it suggests the sample was sufficient to capture the most salient challenges of the clinical workflow.

Ethical Principles

The study adhered to standard ethical guidelines for research involving human participants:

- **Informed Consent:** All participants were provided with a clear description of the study's purpose and gave their explicit consent prior to the interviews and testing sessions.
- **Anonymity and Confidentiality:** To protect the privacy of the healthcare staff, all data was anonymized, and no identifiable personal information was included in the final report.
- **Data Security:** Audio recordings were stored on encrypted drives and were permanently deleted immediately following transcription.
- **Patient Privacy:** The high-fidelity prototype utilized synthetic, non-identifiable patient data to ensure that no real patient information was exposed during the design or evaluation phases.

4. Results

4.1 Identified Usability Challenges

The TA of the seven semi-structured interviews resulted in 108 unique insights, organized under two main thematic pillars: *Work Environment* and *System Interaction.* The analysis revealed a recurring tension between the cognitive demands placed on LPs and CNs and the information available through the current interface. This section presents the primary usability challenges identified across both pillars.

4.1.1 Cognitive Overhead in the Current Interface

A consistent observation across participants was that obtaining a department-level overview through the current spreadsheet-like interface requires substantial cognitive effort. Participants described having to manually aggregate information through mental arithmetic, for example to determine how many patients of a given triage color belonged to a specific team or how many patients had arrived within the last few hours. As one participant noted, simple aggregated measures such as "*how many patients have arrived in the last 1-3 hours"* are not directly accessible and must be computed manually, a task that becomes increasingly burdensome as the patient count grows.

Several participants further described that scrolling becomes a barrier to overview when more than 30 patients are present in the system, and that the interface includes columns perceived as redundant. These observations align with the principles of Cognitive Load Theory [6], where the manual aggregation and visual scanning required by the interface constitute extraneous load that competes with the germane processing needed for clinical decision-making.

4.1.2 Lack of Historical Context

A central finding was that the current interface presents only the *present* state of the department without reference to historical norms. Participants expressed that interpreting the current patient load requires comparing it against an internalized baseline built from years of experience. One participant remarked that they rely on "14 years of experience" while others described forming judgements such as "it is Monday evening, the inflow is this large – good or not good?" based on memory alone.

This pattern was particularly evident among less experienced staff. Participants noted that it takes approximately one year to develop the SA required for the CN role, suggesting that the absence of a visible baseline shifts a significant cognitive burden onto individual expertise. This aligns with Level 2 of Endsley's model [4], where comprehension involves understanding the significance of perceived information. In this setting, that understanding depends on comparing current values against a baseline – which the current interface does not provide.

4.1.3 Fragmented Information and Workflow Disruption

Participants reported that critical information is often distributed across multiple systems and communication channels. Important details such as laboratory results, radiology findings, and bed availability frequently require additional clicks or consultations with colleagues to retrieve. The comment field of the current interface was identified as a particularly problematic locus of information: it contains essential context such as care plans and patient status (e.g *"kakut"* – clear for discharge, *"waiting for transport"*, *"waiting for consultation"*), but only the most recent comment is visible without opening the entry.

This fragmentation forces leadership staff to oscillate between the screen, the clinical floor, and verbal communication with colleagues in order to maintain SA. During periods of high pressure, participants described becoming "stuck behind the computer" for extended periods, sometimes hours, which directly conflicts with the parallel demand of being physically present on the floor.

4.1.4 Absence of Forward-Looking Information

A recurring theme across interviews was that the current overview supports reactive rather than proactive management. Participants described a need for projection – corresponding to Level 3 of Endsley's SA model [4] – but found no support for it in the existing system. One participant explicitly stated that the current overview is "*not fit for purpose for forward planning*" as it conveys only the present state without integrating historical patterns for projected trends.

Concrete examples of unmet projection needs included anticipating staffing requirements for the remainder of a shift, identifying whether the current inflow rate is sustainable, and detecting whether the department is "*catching up or being outrun by the inflow*". These needs map directly to the patient flow dynamics that the proposed dashboard was designed to address.

4.1.5 Summary of Identified Challenges

Synthesizing across the thematic pillars, four primary usability challenges emerged from the analysis:

1. **High extraneous cognitive load** from manual aggregation, scrolling, and visual filtering of a dense spreadsheet-like view.
2. **Lack of historical baselines** to support comprehension of current values.

3. **Information fragmentation** across the interface, requiring multiple clicks and parallel sources to assemble an overview.
4. **Absence of projection support** that would enable proactive rather than reactive department management.

These challenges directly informed the prioritization process described in section 3.3 and shaped the five core dashboard elements presented in section 4.2.

4.2 Dashboard Design

Building on the usability challenges identified in Section 4.1, a high-fidelity dashboard prototype was developed to supplement the current interface. The dashboard was designed to offload extraneous cognitive effort to the system through five core elements arranged according to established principles of visual scanning behavior. This section presents each element and the rationale linking it to the theoretical framework outlined in Chapter 2.

4.2.1 Information Architecture and Layout

The dashboard is organized into a top-level control bar followed by a two-row content area, as shown in Figure 7. The control bar contains two persistent selectors: a team selector that allows the user to filter the view between the overall department and individual teams (A, B, C, X), and a time window selector that adjusts the temporal granularity of relevant elements between 1, 2, and 3 hours, or the current day, week, or month.

The layout is consistent with the F- and Z-pattern scanning behaviors described in section 2.3.3. The most critical real-time metric – the current patient load – is positioned in the upper-left corner of the content area, corresponding to the entry point of both scanning patterns. Categorical breakdowns (age distribution and triage priority) follow horizontally across the upper row, while the temporal trend visualization occupies the lower row in full width. This arrangement ensures that the initial scanning path captures the most decision-critical information first, reducing the time required to reach Level 1 SA [4].

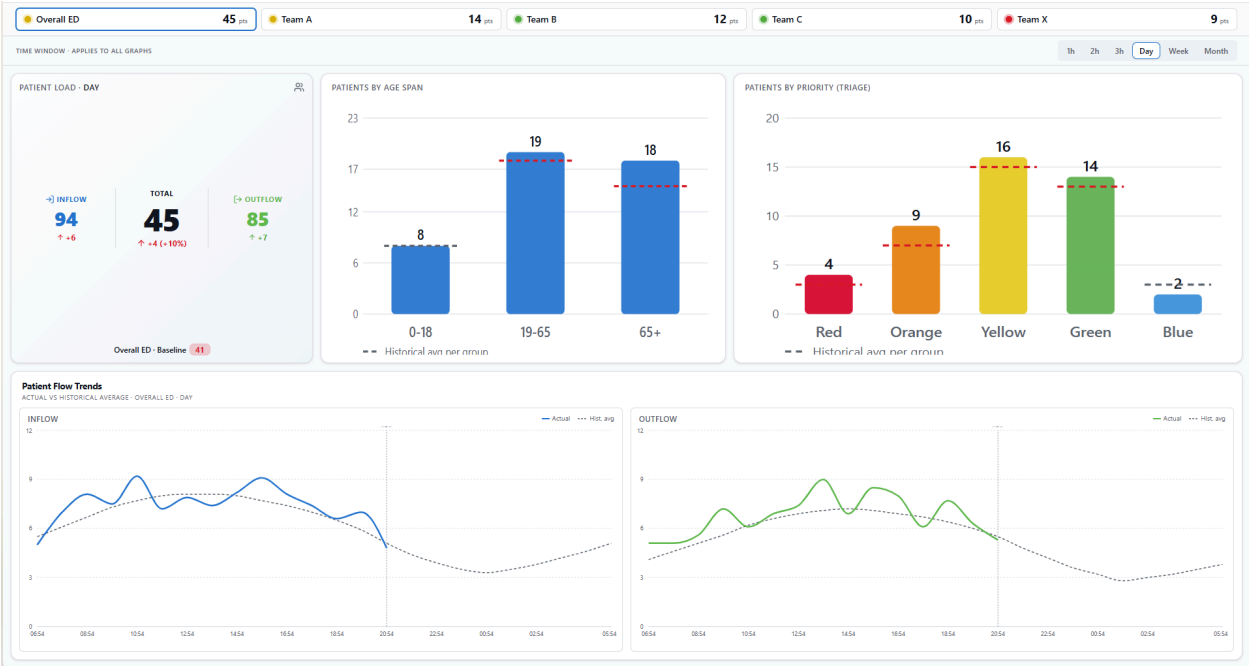


Figure 7. The prototype showing five core elements and time window selectors

4.2.2 Element 1: Patient Load

The Patient Load element presents three key figures: the total number of patients currently in the department, the inflow over the selected time window, and the outflow over the same window. Below each figure, a smaller value indicates the deviation from the baseline for the corresponding interval, color-coded to denote whether the current value is elevated or reduced compared to expected load.

The Total figure remains constant regardless of the selected time window, reflecting the present state of the department, while Inflow and Outflow respond to the time window selector to support both short-term operational and longer-term tactical perspectives. This dual-mode design addresses the recurring need expressed during interviews for aggregated measures that do not require manual computation, such as "*how many patients have arrived in the last 1-3 hours*". By presenting these values alongside their baselines, the element is designed to support Level 2 comprehension as described by Endsley [4], allowing the user to assess whether the current load is normal, elevated, or critical rather than relying on internalized experience.

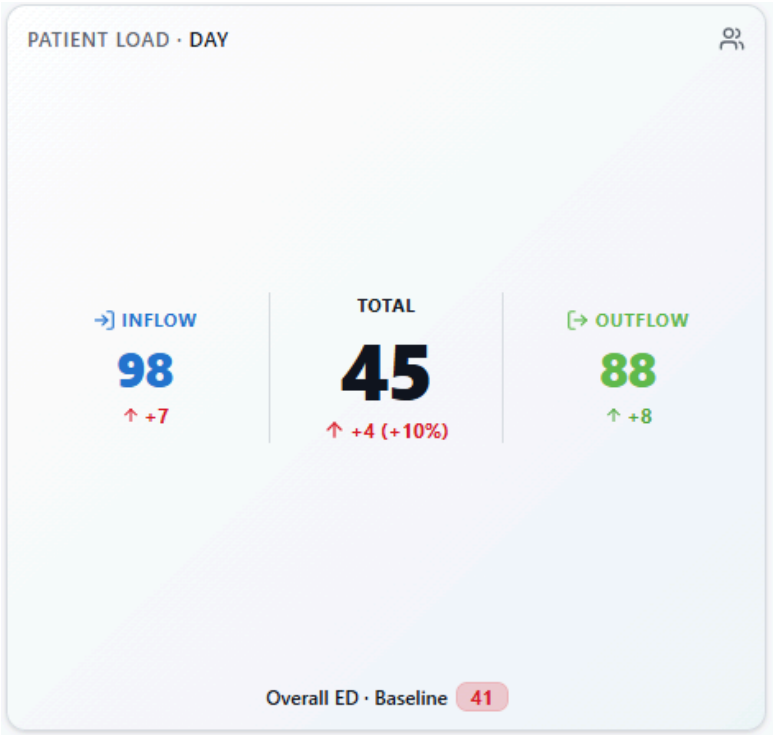


Figure 8. The Patient load overview element

4.2.3 Element 2: Patients by Age Span

The age distribution is presented as a bar chart grouped into three intervals: 0–18, 19–65, and 65+. Each bar is overlaid with a dashed reference line indicating the historical average for that age group during the selected time window. The element was included in response to participants identifying age as a critical data point for clinical assessment, particularly for elderly patients where frailty markedly increases care complexity.

The choice of color-neutral bars for this element was deliberate. As Tufte's principle of data-ink [9] prescribes, decorative variation should be avoided where it does not encode meaning, and as color-coding in the clinical setting is used for triage priority, using additional colors for age would risk introducing semantic conflict. The historical reference line provides the baseline context required to interpret each bar at a glance.

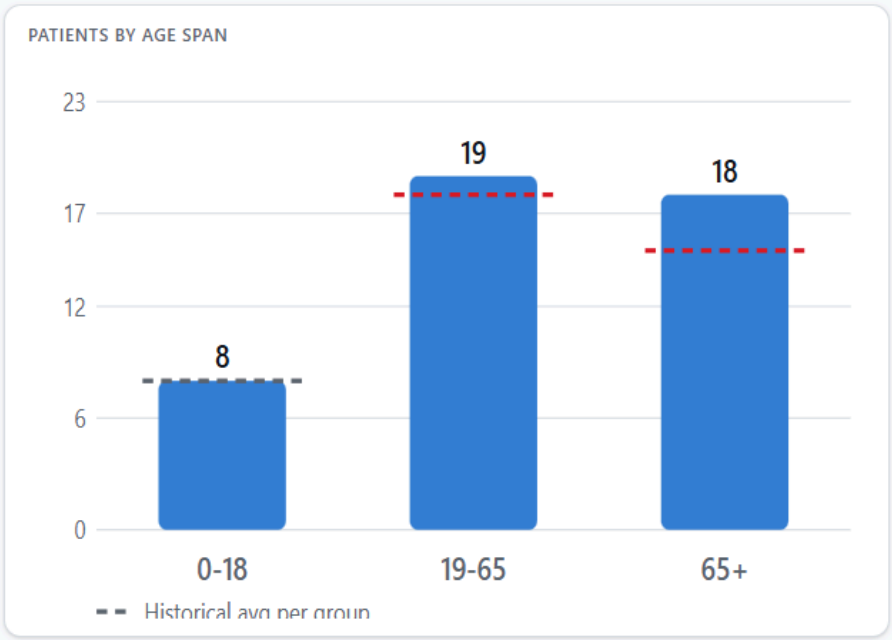


Figure 9. The patient by age span graph element

4.2.4 Element 3: Patients by Priority (Triage)

The triage distribution is presented as a color-coded bar chart following the established medical triage standard: red, orange, yellow, green, and blue. Each bar is overlaid with a dashed historical reference line in the same manner as the age distribution element. This design leverages color as a pre-attentive attribute [7], allowing leadership staff to detect anomalies – for example an elevated number of red or orange patients – before deliberate analysis.

Color-coding by triage priority was the single most frequently cited information requirement across the interview data, consistent with the established role of color as a pre-attentive cue in healthcare interfaces [8]. By preserving the established color semantics from the existing system, the element reduces the learning curve while making the distribution immediately interpretable.

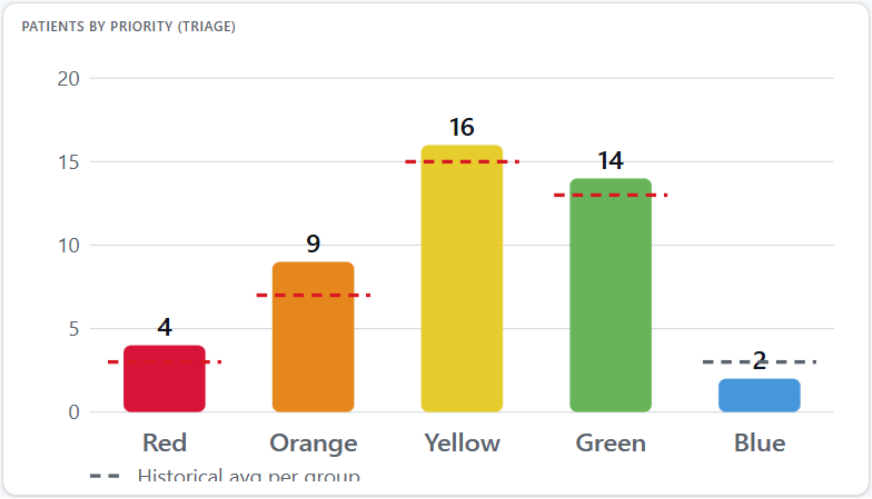


Figure 10. The patient by priority graph element

4.2.5 Element 4: Patient Flow Trends

The Patient Flow Trends element presents two adjacent line charts showing inflow and outflow over the selected time window, each plotted against its historical average. The current time is marked with a vertical reference line, separating observed values from the remainder of the baseline curve.

This element directly addresses the projection gap identified in section 4.1.4. By presenting actual flow against historical norms over time, the element is designed to support Level 3 SA [4], allowing leadership staff to anticipate whether the department is approaching equilibrium or whether a divergence between inflow and outflow signals an emerging bottleneck. The decision to display inflow and outflow as separate but adjacent charts, rather than overlaid in a single chart, was made to preserve the readability of each individual series while still allowing side-by-side comparison.



Figure 11. The patient flow trends element

4.2.6 Element 5: Team-Level Aggregation

The team selector in the control bar provides a fifth dimension to the dashboard by allowing the user to switch between an aggregated overall view and team-specific views. Each team is associated with a composite load indicator displayed alongside the team name in the selector, providing an at-a-glance comparison of relative team load.

This element responds to the organizational structure of the emergency department, where leadership decisions frequently involve redistributing resources between teams or identifying which teams are under elevated pressure. By making team-level comparison accessible without requiring the user to manually filter the spreadsheet-like view, the element reduces the extraneous load associated with this recurring task.



Figure 12. The Team-Level aggregation selector element

4.2.7 Summary of Design Rationale

Table 1 summarizes how each dashboard element addresses the usability challenges identified in section 4.1 and which theoretical principles informed its design.

Table 1. Mapping of dashboard elements to identified usability challenges and their theoretical basis.

Element	Addresses Challenge	Theoretical Basis
Patient Load with baseline	Manual aggregation; Lack of baseline	Cognitive Load Theory [5, 6]; SA Level 2 [4]
Age Span Distribution With baseline	Manual aggregation; Lack of baseline	Data-ink principle [9]; SA Level 2 [4]
Triage Priority Distribution with baseline	Manual aggregation; visual noise	Pre-attentive attributes [7]; color-coding in healthcare [8]; SA Level 2 [4]
Patient Flow Trends	Absence of projection support	SA Level 3 [4]
Team-Level Aggregation	Manual filtering to view team-specific data	Cognitive Load Theory [5, 6]

Together, the five elements form a coherent supplementary view that aims to offload the cognitive aggregation work currently performed mentally by leadership staff, while preserving the data semantics of the current interface.

4.3 Evaluation Through the Technology Acceptance Model

The high-fidelity prototype was evaluated with six clinical participants using a TAM-based survey complemented by open-ended questions and observational notes from the demonstration sessions. The survey comprised 22 scaled items in total; the analysis reported here focuses on the six PU, and six PEOU items, together with three items addressing the specific visualization choices of the dashboard. All items were rated on a seven-point agreement scale ranging from 1 (strongly disagree) to 7 (strongly agree).

4.3.1 Overall TAM Scores

The aggregated results across the two primary TAM constructs are presented in Figure 13. PEOU received a mean rating of M = 6.03 (SD = 0.99), while perceived usefulness received a mean rating of M = 5.25 (SD = 1.06). Both constructs scored above the scale midpoint of 4, indicating overall positive reception of the prototype, with Ease of Use rated higher than Usefulness.

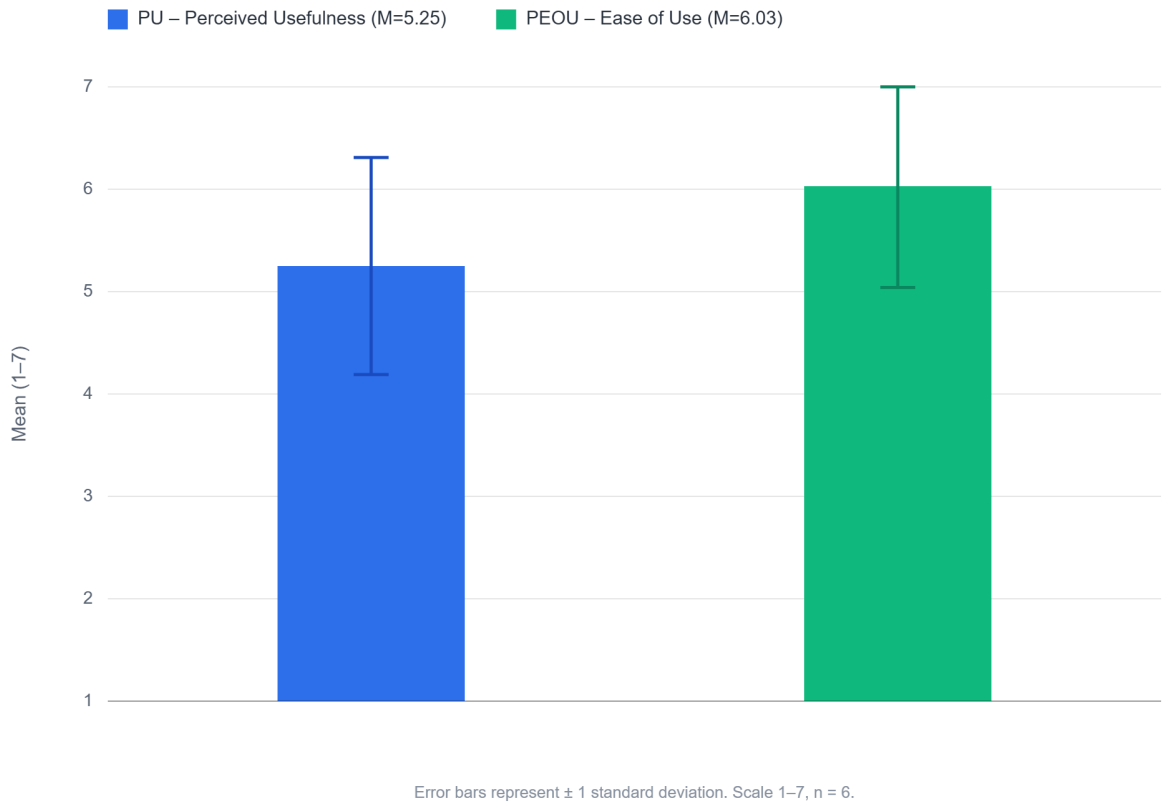


Figure 13. Mean scores for Perceived Usefulness and perceived ease of use. Error bars represent ± 1 standard deviation. Scale 1-7, n = 6.

The higher score for Ease of Use is consistent with feedback gathered during the demonstration sessions, where multiple participants commented on the visual clarity of the interface. One participant described the dashboard as "*clean, light, and easy to overview*", adding that "*This is something Cosmic could learn from"*, while another emphasized the value of "*not having to display all information just because it is*

possible". Clarification: Cosmic (Cambio Cosmic) is the current journal system developed by Cambio, referred to as 'the current interface' throughout this report.

The slightly lower score for PU reflects a more nuanced reception. While participants recognized the potential of the dashboard, several noted that fully assessing its utility would require longer exposure within the actual clinical workflow. As one participant observed, the dashboard "*feels necessary once you have had it for a while*", indicating that the perceived value may increase with sustained use.

4.3.2 Item-level Results

The mean scores and standard deviations for the main twelve TAM items are presented in Figure 14. Within the PU construct, the highest-rated item was "*Using the dashboard in my work would give me a faster overview of the situation in the department*" (M = 5.67, SD = 1.03), while the lowest rated item was "*the dashboard would ease my planning of the department's work*" (M = 4.83, SD = 1.94). The latter item also showed the largest spread of all twelve items, suggesting that perceptions of planning support diverged considerably between participants.

Within the PEOU construct, the highest-rated item was "*it would be quick and smooth for me to learn how the dashboard works*" (M = 6.50, SD = 0.84), and the lowest-rated item was "*I find the dashboard responsive and adapted to my way of working*" (M = 5.00, SD = 1.26). All PEOU items scored at or above the midpoint of 4, with five of six items scoring at or above 6.

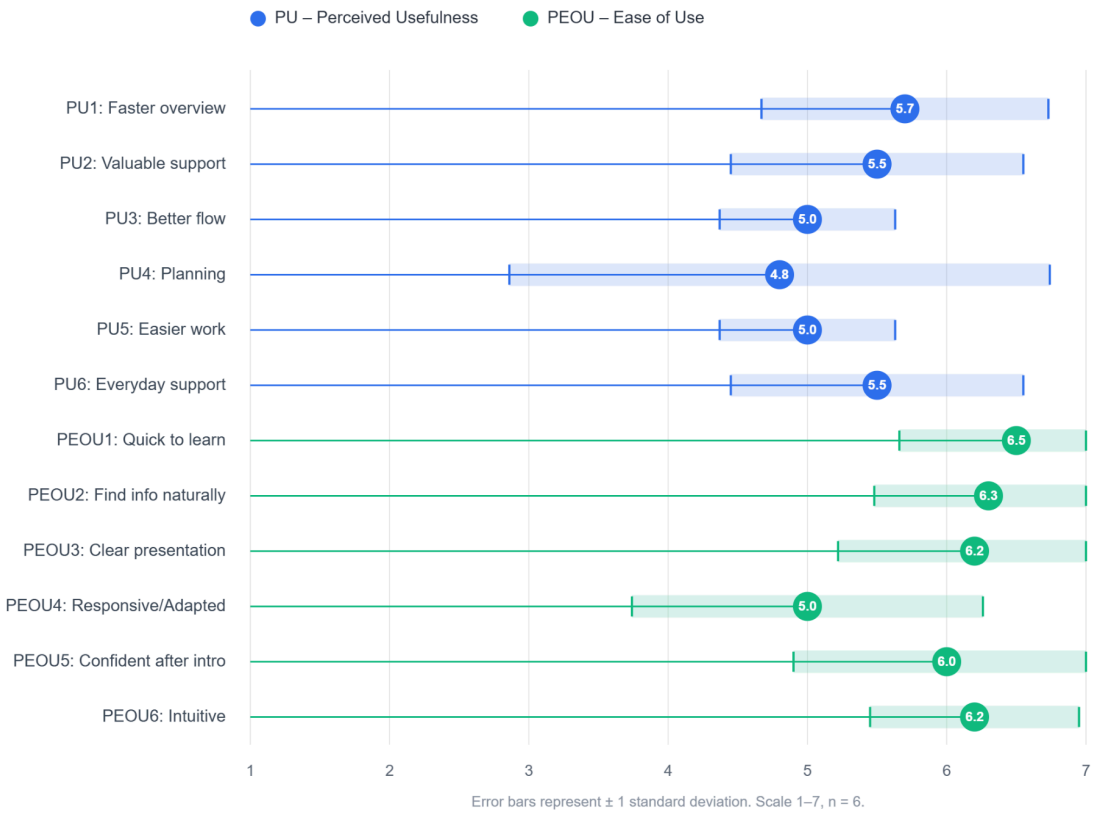


Figure 14. Item-level mean scores and standard deviations for the main twelve TAM items. Color encodes the construct: perceived usefulness(blue) and perceived ease of use(green). Error bars represent ± 1 standard deviation. Scale 1-7, n = 6.

4.3.3 Specific Visualization Items

Three additional items addressed the specific design choices of the dashboard regarding anomaly detection, information density and data trust. The results are presented in table 2.

Table 2. Mean scores and standard deviations for the three specific visualization items.

Item	Mean	SD
I can quickly identify deviations from a normal scenario (e.g. high load).	5.50	1.38
The amount of information displayed feels appropriate.	5.17	1.17
I trust the information shown in the dashboard.	5.67	1.21

The highest score among these items was on data trust, while the appropriateness of information density scored lowest. The comparatively lower score on information density was reflected in qualitative feedback, where one participant noted that "*I do not know what the information is, so it is difficult to know whether to trust it*" – suggesting that trust in displayed values would benefit from greater transparency regarding data sources and calculation logic.

4.3.4 Qualitative Findings on Adoption Factors

The open-ended questions revealed several factors influencing the perceived value of the dashboard and its potential for adoption in daily clinical practice. The feedback was organized around four themes: elements perceived as most valuable, suggested improvements, missing information, and potential adoption barriers.

Most valued elements. When asked what worked best in the prototype, participants highlighted four aspects: the visualization of patient distribution by triage color per team, the overview by priority and age, the historical trend line, and the side-by-side comparison of inflow and outflow against historical averages. One participant summarized the impression as "*relevant information, easy to overview, not too much information, and an interesting prognostic function*".

Suggested improvements. Participants proposed several refinements to the design. These included reordering the triage bars from least to most acute, adding finer-grained age intervals, providing

explanatory text for color signals, and incorporating a summary of patients with abnormally long lengths of stay. One participant proposed introducing a "*patient-hours*" metric to capture cumulative care burden rather than relying solely on patient counts – a measure that could distinguish between high patient volume and high patient complexity.

Missing information. Several participants identified data that they would have valued in the dashboard but found absent. The most prominent suggestion was an indication of patient frailty, particularly for the 65+ age group, with one participant referencing the Clinical Frailty Scale as a relevant measure. Other suggestions included a way of indicating how many patients have been *medically cleared* ("*kakut*") but are waiting on logistical factors such as transport or available beds, and an indicator for whether patients have been reviewed by both nursing and physician staff.

Potential adoption barriers. When asked whether they foresaw any obstacles to introducing the dashboard in daily practice, the responses were largely reassuring. Three participants indicated no perceived barriers, with one noting that "*from a staff perspective it should be easy to introduce*". The principal concern raised related to the validity of the historical baseline against which current values are compared. As one participant noted, the value of the dashboard would be compromised "*if the information it compares against is not current and not updated in real time*". This comment was made in the context of a discussion about using historical data spanning up to twenty years, and reflects an awareness that the emergency department's patient demographics, workflow, and overall load have changed substantially over such a period. The observation underscores the importance of selecting a reference window for the baseline that remains representative of present-day operating conditions, rather than diluting recent trends with outdated historical patterns.

4.3.5 Summary of TAM Evaluation

The TAM evaluation indicates that the proposed dashboard was positively received across both constructs, with notably high scores for Ease of Use and moderately high scores for Usefulness. The qualitative findings further suggest that the perceived value of the dashboard centers on its ability to provide a rapid overview, contextualize current values against historical baselines, and visualize patient flow in a way that supports both retrospective interpretation and forward-looking awareness. The primary factors that would influence broader adoption relate to data freshness, transparency about underlying data sources, and the inclusion of additional clinically meaningful variables such as patient frailty and logistical status.

5. Discussion

This chapter interprets the results presented in Chapter 4 in relation to the three research questions, the theoretical framework established in Chapter 2, and the broader context of designing decision-support interfaces for the emergency care. The chapter first interprets the findings for each research question, then considers their theoretical implications, reflects on the design and the methodological approach, addresses sustainability, ethics and source criticism, and finally outlines directions for future work.

5.1 Interpretation of Findings

5.1.1 RQ1: Usability Challenges in the Current Interface

The TA revealed four primary usability challenges in the current spreadsheet-like interface: high extraneous cognitive load, lack of historical baselines, information fragmentation, and the absence of projection support. These challenges are not isolated technical shortcomings but reflect a deeper mismatch between the design of the current interface and the cognitive demands of the leadership role in an emergency department.

A particularly notable finding was the extent to which experienced staff have compensated for these shortcomings by internalizing reference points over many years of practice. The observation that it takes approximately one year to develop the SA required for the CN role suggests that the current interface effectively externalizes accumulated expertise, but the resulting workflow becomes fragile under high pressure and difficult to transfer to newer staff. This finding aligns with the Cognitive Load Theory, which holds that the way information is presented can impose extraneous load that competes with the germane processing needed for the task [6]. A dense, poorly structured interface therefore consumes cognitive resources that could otherwise support comprehension, which supports the design rationale of shifting aggregation and baseline computation from the user to the system.

5.1.2 RQ2: Visualization of Key Patient Variables

The five core dashboard elements were designed to address each of the identified challenges through evidence-based visualization principles. The TAM evaluation indicates that this approach was largely successful in terms of perceived clarity and ease of use, and the qualitative feedback specifically highlighted the historical trend visualization, the side-by-side comparison of inflow and outflow against historical averages, and the triage distribution per team as the most valuable elements.

A first observation is that several of the dashboard elements directly address the extraneous cognitive load identified as the most basic challenge in Section 4.1.1. The Patient Load element presents the total number of patients, inflow, and outflow as standing figures rather than as values that have to be assembled from a scrollable list of patient rows; the triage and age distribution elements present per-category counts that previously required manual visual filtering of the current interface; and the team-level aggregation provides at-a-glance comparison between teams that previously could only be obtained by mentally scanning across them. Together these elements operationalize the Cognitive Load Theory rationale

outlined in section 2.3.1 – moving the aggregation work from the user to the system -and the consistently high scores on PEOU indicate that this shift was successful in being perceived as effortless to interpret.

The reception of historical baselines was more nuanced. The idea of presenting current values against a historical reference did not emerge as a broadly reported need across the interviews; rather, it was raised primarily by a single participant whose research background is specifically in emergency department patient flow. This gives the element a particular character: it is grounded in deep domain expertise rather than in a widely shared pain point expressed across the leadership group. During the TAM evaluation, participants responded positively to the baseline concept itself, but the feedback also revealed a parallel scepticism regarding the data underlying the baseline and how representative it would be – a concern that was likely amplified by the way the baseline was introduced during the sessions as potentially drawing on historical data spanning up to twenty years. The reception therefore suggests that the value of a historical baseline is recognized in principle, but that its credibility in practice depends heavily on the relevance and transparency of the data behind it.

Where the dashboard does map cleanly onto the theoretical framework is in its layered support for SA. The design choice to display current values alongside historical references operationalizes Endsley's distinction between perception (Level 1 SA) and comprehension (Level 2 SA) [4]. Where the current interface supports Level 1 awareness through raw data, the dashboard adds the reference points required for Level 2 comprehension, and the Patient Flow Trends element extends this support toward Level 3 by enabling projection through the comparison of actual flow against expected flow over time.

A noteworthy observation emerged from comparing the two phases of data collection. During the initial interviews, participants found it difficult to articulate concrete needs when asked to envision decision-support features in the abstract without an actual prototype in front of them. In contrast, when presented with the high-fidelity prototype during the TAM evaluation, the same category of users readily generated a substantial number of concrete and clinically relevant suggestions. This contrast suggests that users are considerably better at reacting to a tangible artifact than at imagining, which is consistent with the rationale behind prototyping as a stage in the Design Thinking Process [2]. It also implies that the five core elements, while grounded in the interview data, are best understood as a well-founded starting point rather than a definitive set; a further design iteration informed by the prototype-driven feedback would likely refine them further.

5.1.3 RQ3: Perceived Acceptance and Factors Influencing Adoption

The TAM evaluation showed that both PU ($M = 5.25$, $SD = 1.06$) and PEOU ($M = 6.03$, $SD = 0.99$) scored well above the scale midpoint, indicating overall positive reception. The higher score for Ease of Use compared to Usefulness is consistent with the pattern reported by Davis [14], where ease of use exerted a direct effect on usage immediately after a brief introduction but not after fourteen weeks of use, while usefulness remained the dominant predictor throughout – suggesting that ease of use weighs more heavily in early impressions, whereas usefulness becomes decisive with sustained use. The qualitative finding that the dashboard "*feels necessary once you have had it for a while*" supports this interpretation: the full utility of the system may only become apparent in sustained clinical practice.

The lowest-scoring PU item – concerning whether the dashboard would ease the planning of the department's work – also exhibited the largest spread of all twelve items (SD = 1.94). This divergence is consistent with an observation raised during the initial interviews, where it was noted that the leadership roles tend to be oriented toward the operational here-and-now rather than toward forward planning. At the same time, the data collection also included participants whose perspective was clearly oriented toward the forward-looking and strategic side of the work, including a participant with a research background in emergency department patient flow and a participant holding a senior clinical leadership role beyond the day-to-day LP and CN functions. Both orientations were therefore represented across the data, and the contrast between them offers a tangible basis for interpreting the spread on the planning item. While the dashboard supports both an aggregated overview of the current situation and a forward-looking dimension through the comparison of actual flow against historical averages, the planning-related item specifically targets the latter. The wide spread on this item can be read in that light: participants whose work involves a planning component may have engaged readily with the forward-looking aspect, while those focused predominantly on operational tasks were less certain of its relevance to their day-to-day work.

The participants in the TAM evaluation were all recruited on the basis of having a direct working relationship with the current interface at the emergency department, ensuring that the feedback came from relevant domain users rather than from staff without exposure to the system the dashboard is intended to supplement. However, the survey item used to capture professional role mixed leadership function and professional role as response options, which means the responses do not reliably distinguish which participants held a formal leadership role – LP or CN – at the time of evaluation. As a result, the qualitative interpretation offered above, which draws on observations made during the interview phase, cannot be complemented by a quantitative role-based analysis of the TAM scores themselves. A more precisely structured role item, combined with a larger sample, would allow the relationship between role orientation and perceived usefulness to be examined more directly in the future evaluations.

5.2 Theoretical Implications

The results provide qualitative support for the theoretical framework established in Chapter 2. The strong reception of pre-attentive encoding through color and position aligns with the established literature on information visualization in healthcare [7,8], and the consistent emphasis on triage color as one of the most frequently cited information requirements reinforces the role of color as a pre-attentive cue in clinical settings. Endsley's levels of SA [4] also held up well as a design framework: the dashboard's separation between current values, historical references, and projected flow correspond closely to the distinction between perception, comprehension, and projection, and participants' positive reception of the layered presentation suggests that the framework captures something real about how clinicians construct an understanding of the department's state.

At the same time, the results illustrate a recognized limitation of the Technology Acceptance Model itself. TAM measures intention to use rather than actual use, and high intention scores do not necessarily translate into adoption in practice [24]. This gap is particularly relevant in healthcare, where organizational, infrastructural, and cultural factors mediate the path from acceptance to implementation. The qualitative findings of this study point in the same direction: the concern about baseline validity, the requests for additional clinical variables, and the implicit dependence on integration with live data all

indicate that adoption would rest on factors well beyond PU and PEOU. TAM was therefore valuable as a structured way to assess initial reception, but it is best used as one input into a broader adoption picture rather than as a prediction of clinical uptake.

A further observation concerns the relationship between the two TAM constructs. Davis [14] found that ease of use influences usage primarily indirectly, by way of perceived usefulness, and that usefulness is the stronger determinant of adoption. The present results are consistent with this hierarchy: ease of use scored highly and was readily assessed after a short demonstration, while usefulness scored lower and was repeatedly described as something that would become clearer with sustained use. The pattern reinforces Davis's argument that a system which is easy to use but not yet perceived as clearly useful is not yet positioned for adoption – and that the path toward adoption runs through demonstrating usefulness in real clinical practice.

5.3 Design Reflections

While the TAM evaluation provides an overall positive picture of the dashboard, the qualitative feedback and the design process itself surface a number of more specific reflections that are worth addressing on their own.

5.3.1 Information Density

Among the three specific visualization items in the survey, the appropriateness of information density received the lowest score (M = 5.17, SD = 1.17), and the qualitative feedback included both requests for additional data – such as patient frailty, logistical status, and length-of-stay outliers – and requests for clearer explanations of existing elements. This points to a tension that is inherent in the design of decision-support interfaces: minimizing extraneous load argues for sparsity, while supporting comprehensive SA argues for completeness. The current prototype leans toward sparsity, and the feedback suggests that a small number of carefully chosen additions would strengthen the perceived utility without compromising clarity. These include, in particular, a frailty indicator and a logistical-status indicator, that is, a way of showing how many patients are medically cleared but still occupying space while awaiting transport, beds, or other non-medical resolution.

5.3.2 The Team-Level Indicator as a Minimal Overview Solution

During the design phase it became apparent that some form of cross-sectional overview could be valuable – both a lightweight sense, from any view, of how other parts of the department are doing, and potentially a dedicated view presenting all teams at once. It was not clear how to provide this without making the interface cluttered, which is itself an instance of the information-density tension discussed in Section 5.3.1. The design settled on a deliberately minimal compromise: a status indicator attached to each team's selector, giving a rough sense of a team's situation without requiring a switch of view. This solution addresses only the lightweight version of the need and leaves an open question that the prototype does not resolve – namely how the underlying assessment behind such an indicator should be calculated and, importantly, how its basis should be made transparent to the user. This is taken up further in Sections 5.7.4 and, in connection with data validity, 5.5.2.

5.3.3 Ordering of the Triage Bars

A specific suggestion from the evaluation was to reorder the triage bars from least to most acute, rather than from most acute to least acute as in the current prototype. The current ordering follows the same left-to-right scanning logic that informs the overall layout of the dashboard, as discussed in section 2.3.3: placing the most acute category at the entry point of the user's scanning path makes the most safety-critical information the first to be visually registered. However, at least one participant read the chart through a different convention – that of a numerical scale where increasing acuity should appear at the far end – and interpreted the ordering accordingly. This is a small detail but illustrative of a broader observation: visual orderings can sit at the intersection of more than one convention, and an interface that selects one without explanation cannot rely on it being read uniformly across users.

5.3.4 Placement of Triage Relative to Scanning Patterns

A related observation concerns the position of the triage distribution within the layout. Triage color was, in the interview data, one of the most frequently cited information requirements, yet in the current prototype it is placed in the upper-right rather than the upper-left of the content area. The upper-left position is occupied by the Patient Load element, which gives a numerical summary of the department as a whole – how many patients are present, and how many have arrived and left over the selected time window. The choice can be defended on functional grounds: the categorical breakdowns of triage and age are not fully meaningful in isolation, but acquire their meaning in relation to the total. Ten patients distributed across acuity categories represents a fundamentally different situation than eighty patients with the same distribution, even though the proportions between the bars would look the same in both cases. Patient Load therefore acts as the reference against which the categorical breakdowns can be read, and placing it first establishes that reference before the user moves on to the breakdowns themselves.

5.3.5 System Familiarity

Several participants reflected during the evaluation that their long experience with the current interface made it difficult to assess a new interface in fully neutral terms. This was a recurring theme across the evaluation sessions: users develop perceptual shortcuts in the systems they are habituated to, and judging an alternative requires temporarily setting those shortcuts aside. The implication of the present results is that participants brought their current interface familiarity with them into the evaluation, which makes it difficult to know whether the TAM scores reflect the dashboard on its own terms or whether that familiarity influences the responses. A longer period of exposure would be needed before the dashboard could be assessed independently of that familiarity, and additional evaluation methods that do not rely on the participants' prior experience with related systems could help establish a more independent baseline.

5.3.6 Scope in Relation to the Wider Information Environment

The interviews surfaced a broader information landscape in which leadership staff regularly draw on sources outside the current interface, such as laboratory results, radiology findings, and bed availability. The present work has deliberately focused on the 'unit overview' section of the current interface itself, as defined in section 1.1, and the dashboard correspondingly addresses the data that originates there. Improvements to the wider information environment fall outside this scope, and any future system aimed

to consolidate clinical information across sources would represent a substantially larger integration effort than the one undertaken here.

5.4 Methodological Reflections and Limitations

This section reflects critically on the methodological choices made throughout the project and the limitations they impose on the interpretation of the results. The reflections are grouped into limitations relating to the sample and evaluation context, to the data collection and analysis process, to the research approach and tooling, and to the framing of the project.

5.4.1 Sample and Evaluation Context

Seven participants took part in the interviews during the Empathize phase, and six participated in the TAM evaluation, out of a population of approximately fifty to sixty LPs and CNs at the emergency department. As discussed in Section 3.6, this sample size is modest but consistent with saturation research on the emergence of the most salient themes [18]; the quantitative TAM results, however, should be interpreted as indicative rather than statistically generalizable. The participants in the interview phase and the evaluation phase were largely different individuals, with one – the clinical academic supervisor – taking part in both phases. This means the design was tested with a partly different group than the one whose needs informed it, and that one participant with a clearly forward-looking orientation contributed to both data sets.

The TAM evaluation was also conducted immediately following a demonstration of the prototype rather than after sustained use in clinical practice. Several participants explicitly noted that the full value of the dashboard could only be assessed after prolonged exposure, and the scores in section 4.3 therefore reflect first-impression acceptance rather than longitudinal adoption. In addition, the survey item used to capture professional role mixed leadership function with professional profession as response options, which limited the ability to analyze how perceptions varied by role; this is discussed in section 5.1.3 and would be straightforward to correct in a future evaluation with a more precisely structured instrument.

A further limitation concerns the background items in the evaluation survey. The item on experience asked for years in the LP or CN role rather than years of experience working with the current interface; the latter would have been more directly relevant to the system the dashboard is intended to supplement, since familiarity with that interface – rather than tenure in the role as such – is what shapes the comparison a participant can make.

5.4.2 Data Collection and Analysis

The researchers had limited prior experience with both qualitative interviewing and interface design. The interview technique was refined iteratively across the seven sessions, as described in section 3.1.2; while this improved the quality of later interviews, it also means that the earliest interviews may have yielded less rich data than the later ones, and that themes identified later in the process could not be probed retrospectively with the first participants. The consultation of a user experience and user interface coach and the project supervisor provided external validation of the design direction, but the final design choices

nevertheless reflect the researchers' level of experience, particularly in areas such as information density and visual hierarchy where established heuristics could have been applied more systematically.
The interview audio was transcribed using an automated speech-recognition model, and the transcripts were subsequently reviewed manually. For the analysis of the TAM evaluation sessions, the transcripts were used without the accompanying audio, which means that some interpretation of what was said relied on the researchers' recollection of the sessions rather than on the recording alone. The analysis of data from the initial interviews followed an inductive approach, in which the goal was not to confirm a predetermined theory but to let the data surface its own themes; this mitigates some forms of confirmation bias but does not eliminate them, since the researchers' attention during extraction was inevitably shaped by their prior exposure to the domain.

A further reflection concerns the framing of the evaluation sessions themselves. As noted during the early consultation, clinical staff tend to engage with a prototype in practical terms, focusing on concrete faults rather than on the underlying concept being tested. This meant that the way each session was introduced – orienting participants toward assessing whether the type of data and visualization could provide value, rather than toward critiquing the prototype as a finished product – had a material influence on the quality of the responses. The introduction was refined over the early sessions as the researchers became aware of this effect, which means that, as with the interview technique, the framing was not fully consistent across all participants. A more carefully scripted and piloted introduction, established before data collection began, would have reduced this source of variation.

A specific procedural change should also be noted. The original interface presented team selection through a dropdown alone. This was subsequently extended with dedicated per-team buttons while the dropdown was retained, but no equivalent button was provided for returning to the department-wide overview. During the first two evaluation sessions it was observed that participants struggled to navigate back to the overall view, repeatedly asking how to return to it. In response, the dropdown was removed after the first two sessions and replaced with an explicit control for the department-wide view, resulting in the layout used for the remaining sessions. This means the first two participants evaluated a version of the prototype with a known navigation problem that the subsequent participants did not encounter, which introduces a small inconsistency into the evaluation data that the reported standard deviations do not capture. At the same time, the episode is itself a minor usability finding: the difficulty of returning to the overview surfaced directly through observation during the sessions and was resolved before it could affect the remaining participants.

5.4.3 Research Approach and Tooling

The research process followed an abductive logic, in which design decisions and methodological choices emerged iteratively from engagement with the problem domain, and the theoretical framework was consulted in parallel to ground and explain the choices that arose. This approach is consistent with established practice in design research and qualitative inquiry, where the relevance of specific theories cannot always be determined in advance of empirical engagement. It does, however, carry a risk of confirmation bias, in that supporting literature may be more readily identified than dissenting literature once a direction has been chosen.

AI-based tools were used as a support throughout the project. During the design and analysis phases, such tools were used to accelerate iteration on the prototype and to support reflection during the synthesis of themes; the high-fidelity prototype itself was generated using an AI-driven prototyping platform, from a build specification that was in turn drafted with AI assistance from the core elements the authors had identified (Section 3.4). During the writing phase, AI-based tools were used to assist with the formulation of academic prose, while the underlying findings, reflections, and judgements remained the work of the authors, who first articulated the substantive content and structure of each section before any linguistic refinement. AI-based tools were also used to produce the recreation of the current interface shown in Figure 1, populated with fictitious data; the authors verified its layout and column structure against the actual system. This way of working enables rapid iteration but limits reproducibility in two respects: the contribution of AI-based tools to specific design decisions and phrasings is not individually traceable, and the same applies to the prototype, whose construction through an AI-driven platform is less transparent than a conventionally documented implementation would be.

The use of an AI-driven prototyping platform also raises a question separate from reproducibility: who owns the resulting code. Lovable's terms state that the platform does not claim ownership of the code it generates and that this rests with the user [25]. Whether the code is protected at all is a further and less settled question: under EU and Swedish copyright law, protection requires a work to be a human author's own intellectual creation reflecting free and creative choices [26], and output generated largely autonomously from a prompt may not meet this threshold, leaving it in a legal gray area [26]. For the present work this has no practical bearing: the prototype is a research artifact used to convey a design direction to users, not a deployed or commercialized product, and its contribution to the thesis lies in the design rationale and the evaluation rather than in the code. The question would become material only if the design were carried toward a real, owned system, as outlined in section 5.7, at which point the intellectual-property status of generatively produced code would need deliberate handling.

Finally, the same two researchers conducted both the design and the evaluation of the prototype. While analyst triangulation during the TA mitigated bias in the qualitative findings, the dual role inevitably introduces the possibility that the evaluation was disposed favorably toward design choices the researchers had already committed to.

5.4.4 Framing of the Project

The direction of the project was shaped to a notable extent by an early consultation with the clinical academic supervisor, who has a research background in emergency department patient flow. This provided the project with deep domain expertise and clear analytical entry points, but it also means that the framing of the problem – what counts as a relevant pain point, and what kind of solution is worth pursuing – carried that perspective into the empirical work. The subsequent interviews and the prototype evaluation were used to test the resulting ideas with a broader group of users, and the generally positive reception suggests that the framing was not idiosyncratic. Nonetheless, the influence of a single expert perspective on the project's initial direction should be acknowledged as a factor that shaped which problems were investigated and which were not.

5.5 Sustainability and Ethical Considerations

5.5.1 Sustainability

The most relevant sustainability dimension for this work is social rather than ecological. The leadership roles in an emergency department operate under sustained cognitive pressure, and the interviews indicated that maintaining an overview of the department currently depends heavily on mental aggregation, memory, and accumulated experience. A tool that shifts part of this work from the individual to the system addresses a genuine working-environment concern: reducing avoidable cognitive load in a high-pressure clinical role is relevant not only to performance but to the well-being of the staff who occupy that role. While this study did not measure stress or workload directly, the design rationale and the positive reception of the dashboard's clarity suggest that a tool of this kind could contribute to a more manageable working situation for leadership staff.

A related social dimension concerns accessibility of the role itself. The interviews indicated that it takes approximately one year to develop the SA required for the CN role, much of it through internalized experience. By making historical context and aggregated overview explicit in the interface rather than dependent on memory, a tool of this kind could lower the threshold for less experienced staff to perform the role competently, which has implications for resilience and continuity in a role where staffing pressure is a recurring concern.

From an economic and organizational perspective, clearer overview and earlier detection of emerging bottlenecks could support more effective use of the limited resources available in an emergency department. One participant highlighted that organizational decision-making often relies on patient counts, while the actual burden is better captured by the time patients spend in the department; a tool that makes such measures visible could support more accurate resource allocation and a better-grounded basis for organizational decisions.

These potential benefits should, however, be stated with appropriate restraint. The dashboard is a decision-support tool: it helps leadership staff perceive and interpret the state of the department, but it does not by itself resolve the structural conditions – staffing levels, bed capacity, patient inflow – that generate cognitive load in the first place. Its sustainability contribution is therefore best understood as supporting the people who manage these conditions, rather than as a solution to the conditions themselves.

5.5.2 Ethical Considerations

The research was conducted in accordance with the ethical principles described in Section 3.6, including informed consent, voluntary participation, anonymity, secure handling of recordings, and the exclusive use of synthetic, non-identifiable patient data in the prototype. These measures addressed the principal ethical risks associated with involving clinical staff as participants and with working in a setting where real patient data is present.

Two further ethical considerations are specific to the nature of this work. The first concerns the use of AI-based tools, discussed methodologically in section 5.4.3. Beyond reproducibility, transparency about such use is itself an ethical matter: the authors have therefore chosen to state the role of these tools openly rather than leave it implicit, on the principle that the basis on which academic work is produced should be disclosed. Relatedly, the authors retain full responsibility for the report's content. AI assistance language was applied only to phrasing whose substance the authors had already determined, and all factual statements, citations, and reported data were verified by the authors; no claim or reference was included that they had not independently checked and could not stand behind.

The second concerns the responsibility carried by a decision-support tool in a patient-critical environment. A dashboard that presents current values against a historical baseline implicitly invites the user to trust that baseline. If the underlying data is unrepresentative or outdated – a concern raised directly during the evaluation – the tool could foster a false sense of confidence rather than genuine SA. This places an ethical requirement on any future development: the provenance and limitations of the data must be transparent to the user, so that the dashboard supports clinical judgement rather than quietly substituting for it. This consideration is not unique to the present prototype but applies to any system that mediates SA in a clinical setting.

5.6 Source Criticism

The sources used in this work are, for the most part, established and widely cited, but a few characteristics of the reference base warrant comment.

A number of the sources that inform the methodological and theoretical framing are practitioner-oriented rather than peer-reviewed. Several of the references underpinning the design process and the visual scanning behaviors originate from professional design organizations rather than from primary research literature. These sources are well regarded within the field of interaction design and are appropriate for conveying established design practice, but they do not carry the same evidentiary weight as peer-reviewed studies. Where such sources support load-bearing parts of the argument – in particular the scanning patterns used to motivate the dashboard layout – the conclusion drawn from them should be read as grounded in accepted design practice rather than in primary empirical evidence, and the layout decisions they informed are ones that this study explicitly identifies as warranting further empirical testing.

A second observation concerns the age of several core sources. A number of the foundational works drawn on here – for cognitive load, SA, technology acceptance, and information design – date from the 1980s and 1990s. This is not in itself a weakness: these are seminal works that continue to be cited and that define the constructs they introduced. However, in several cases the original source was used rather than the more recent literature that has extended or qualified it. The Technology Acceptance Model is the clearest example: while the original formulation provides the constructs measured in this study, later work has examined its limitations in healthcare setting specifically, and the present work engages with that critique only briefly. The reliance on foundational sources therefore provides a solid conceptual basis but does not fully reflect how these frameworks have developed since.

Finally, as discussed in section 5.4.3, literature was to a large extent consulted in parallel with or after design and methodological decisions had been made, rather than driving from the outset. This carries the risk that supporting sources were more readily identified than dissenting ones, and the reference base should be read with that selection effect in mind.

5.7 Future Work

This chapter outlines directions for future development that emerged from the evaluation findings and the limitations of the current study. The identified areas span refinements to existing functionality, additions of new clinical variables, and methodological extensions to the evaluation itself.

5.7.1 Baseline Computation

A central refinement identified through the evaluation concerns how the historical baseline is computed, as it relates directly to the principal adoption concern raised by participants. That concern was not the baseline concept itself but the relevance of the data behind it: the risk that a baseline drawn from too long a history no longer represents current conditions, as older data that no longer reflects the department's reality is averaged together with recent patterns. Future work should therefore investigate methods for constructing a baseline that reflects current operating conditions, such as moving averages, day-of-week and time-of-day segmentation, and decay-weighted historical data that gives greater weight to recent periods. Resolving this is central to the dashboard's support for Level 2 SA, since the value of a reference point depends entirely on its representativeness.

5.7.2 Empirical Analysis of Historical Data

A complementary line of work would involve an empirical analysis of historical data from the emergency department itself. Such an analysis would serve two purposes: it would inform the design of the baseline computations described above, and it could surface previously unrecognized patterns in patient flow, length of stay, and triage distribution that are clinically meaningful in their own right. This would also help establish which historical window is genuinely representative, addressing the validity concern on an empirical, rather than an assumed basis.

5.7.3 Additional Clinical Variables

Participants identified several clinical variables that would strengthen the dashboard's utility but were not part of the current prototype. These included some indication of patient frailty as a measure of care burden, for instance drawing on an established measure such as the Clinical Frailty Scale; a way of indicating how many patients are medically cleared but still occupying space while awaiting logistical resolution such as transport or available beds; and a metric capturing the cumulative time patients have spent in the department, which would distinguish high patient volume from high care burden. A related refinement of this last measure, raised during the evaluation, would be to surface the extent to which a

small number of unusually long-staying patients account for a disproportionate share of the department's total patient time, since such outliers can dominate the time burden without being visible in patient counts alone. Each of these would need to be introduced carefully in light of the information-density tension discussed in section 5.3.1, but the fact that they were raised independently by participants suggests they would meaningfully extend the dashboard's clinical relevance.

5.7.4 Cross-Sectional Overview Across Teams

During the design phase it was anticipated that some form of cross-sectional overview could be valuable: both a lightweight sense, from any view, of how the other parts of the department are doing, and a dedicated view presenting all teams individually and simultaneously. It was not clear how to provide this without making the interface cluttered, and the design settled on a compromise: a status indicator attached to each team's selector, giving a rough sense of each team's situation without a dedicated view. This addresses only the lightweight version of the need and was a deliberately minimal solution. The need for fuller cross-sectional overview was subsequently expressed by participants during the evaluation, which provides empirical support for an intuition that had until then only informed a minimal prototype solution. Future work should therefore explore how a dedicated multi-team overview is best designed so that all teams can be viewed in parallel without overwhelming the user, and – as part of that – how the underlying assessment behind any such indicator is calculated and made transparent, in line with the broader data-validity concern discussed in sections 5.1.3 and 5.5.2.

5.7.5 Automated Flagging and Decision Support

Several participants expressed interest in features that go beyond passive visualization, such as automatic flagging of patients who have remained in the department beyond expected thresholds. Such features open the possibility of incorporating machine learning components in future iterations to automate routine monitoring tasks that currently depend on manual attention. Any such development would, however, raise the same data-transparency requirement discussed in section 5.5.2: an automated system that contributes to SA carries a heightened responsibility to make the basis of its outputs visible to the clinician.

5.7.6 Longitudinal Evaluation

Finally, the most significant methodological extension would be a longitudinal evaluation in which the dashboard, or an extended version of it, is deployed in the clinical environment over a sustained period. This would address the central limitation of the present evaluation, namely that perceived usefulness was assessed on the basis of a short demonstration rather than real use. A longitudinal study would allow PU to be reassessed once the dashboard had been incorporated into actual workflow, would reduce the influence of system familiarity discussed in section 5.3.5, and – if combined with a more precisely structured role item – would permit the role-based analysis that the present dataset could not support.

6. Conclusion

This thesis set out to investigate how alternative visualization methods and interface design principles could improve the presentation of patient information in the emergency care unit at Linköping University Hospital, with a focus on supporting clearer overview, improved prioritization, and better usability for lead physicians and chargenurses.

Addressing the first research question, a thematic analysis of interviews with clinical leadership identified four recurring usability challenges in the current spreadsheet-like interface: a high extraneous cognitive load arising from manual aggregation, an absence of historical context against which current values can be interpreted, fragmentation of information across systems and channels, and a lack of support for forward-looking assessment. These challenges were found to stem less from missing data than from how information is structured and presented.

In response to the second research question, a high-fidelity dashboard prototype was developed in which five core elements were arranged according to established principles of cognitive load, situation awareness, and visual scanning behavior. The design aimed to shift aggregation work from the user to the system and to make historical context explicit rather than dependent on individual experience.

For the third research question, the prototype was evaluated using a survey based on the Technology Acceptance Model, complemented by qualitative feedback. The dashboard was received positively with notably high scores for perceived ease of use and moderately high scores for perceived usefulness. The qualitative findings indicated that the perceived value centered on rapid overview, the contextualization of current values against historical baselines, and the visualization of patient flow, while the factors most likely to influence broader adoption concerned the validity and transparency of the underlying data rather than the interface itself.

Taken together, the results suggest that a theoretically grounded dashboard can address usability challenges that clinical leadership currently manages through experience and mental effort, and that such an approach is, in principle, well received by the intended users. These conclusions should, however, be read as indicative rather than definitive. The work is exploratory and was conducted with a small sample, a short evaluation horizon, and a prototype rather than a deployed system. Its primary contribution is therefore not a finished solution but a grounded foundation: a set of identified needs, a design rationale linking those needs to established theory, and an initial indication that the approach merits further development. The directions outlined in section 5.7 – in particular a representative baseline computation, additional clinical variables, and a longitudinal evaluation in real clinical use – represent the natural next steps toward determining whether the value indicated here holds under sustained practical conditions.

References

[1] *Ergonomics of Human-System Interaction - Part 210: Human-Centered Design for Interactive Systems*, ISO 9241-210:2019, International Organization for Standardization, Geneva, Switzerland, 2019.

[2] Interaction Design Foundation, "The 5 stages in the design thinking process," 2026. [Online]. Available: https://ixdf.org/literature/article/5-stages-in-the-design-thinking-process. [Accessed: Apr. 20, 2026].

[3] Interaction Design Foundation, "What is creative problem solving?" Jun. 14, 2024. [Online]. Available: https://ixdf.org/literature/topics/creative-problem-solving. [Accessed: Aug. 28, 2026].

[4] M. R. Endsley, "Toward a theory of situation awareness in dynamic systems," *Human Factors*, vol. 37, no. 1, pp. 32-64, 1995, doi: 10.1518/001872095779049543.

[5] J. Sweller, "Cognitive load during problem solving: Effects on learning," *Cognitive Science*, vol. 12, no. 2, pp. 257-285, 1988, doi: 10.1207/s15516709cog1202_4.

[6] J. Sweller, J. J. G. van Merriënboer, and F. G. W. C. Paas, "Cognitive architecture and instructional design," *Educational Psychology Review*, vol. 10, no. 3, pp. 251-296, 1998, doi: 10.1023/A:1022193728205.

[7] A. Treisman, "Preattentive processing in vision," *Computer Vision, Graphics, and Image Processing*, vol. 31, no. 2, pp. 156-177, 1985.

[8] R. Fusco et al., "Visual perception and pre-attentive attributes in oncological data visualisation," *Bioengineering*, vol. 12, no. 7, art. no. 782, Jul. 2025, doi: 10.3390/bioengineering12070782.

[9] E. R. Tufte, The Visual Display of Quantitative Information. Cheshire, CT, USA: Graphics Press, 1983.

[10] J. Nielsen, "F-shaped pattern for reading web content," Nielsen Norman Group, Apr. 16, 2006. [Online]. Available: https://www.nngroup.com/articles/f-shaped-pattern-reading-web-content-discovered/. [Accessed: Jul. 20, 2026].

[11] M. Soegaard, "Visual hierarchy: Organizing content to follow natural eye movement patterns," Interaction Design Foundation. [Online]. Available: https://ixdf.org/literature/article/visual-hierarchy-organizing-content-to-follow-natural-eye-movement-patterns. [Accessed: Jul. 20, 2026].

[12] V. Braun and V. Clarke, "Using thematic analysis in psychology," *Qualitative Research in Psychology*, vol. 3, no. 2, pp. 77-101, 2006, doi: 10.1191/1478088706qp063oa.

[13] V. H. Vroom, Work and Motivation. New York, NY, USA: Wiley, 1964.

[14] F. D. Davis, "Perceived usefulness, perceived ease of use, and user acceptance of information technology," *MIS Quarterly*, vol. 13, no. 3, pp. 319-340, 1989, doi: 10.2307/249008.

[15] F. D. Davis, R. P. Bagozzi, and P. R. Warshaw, "User acceptance of computer technology: A comparison of two theoretical models," *Management Science*, vol. 35, no. 8, pp. 982-1003, Aug. 1989, doi: 10.1287/mnsc.35.8.982.

[16] A. Bandura, "Self-efficacy mechanism in human agency," *American Psychologist*, vol. 37, no. 2, pp. 122-147, 1982, doi: 10.1037/0003-066X.37.2.122.

[17] M. Q. Patton, *Qualitative Research & Evaluation Methods: Integrating Theory and Practice*, 4th ed. Thousand Oaks, CA, USA: SAGE Publications, 2014.

[18] G. Guest, A. Bunce, and L. Johnson, "How many interviews are enough? An experiment with data saturation and variability," *Field Methods*, vol. 18, no. 1, pp. 59-82, 2006, doi: 10.1177/1525822X05279903.

[19] S. Kvale and S. Brinkmann, *InterViews: Learning the Craft of Qualitative Research Interviewing*, 2nd ed. Thousand Oaks, CA, USA: SAGE Publications, 2009.

[20] J. C. Flanagan, "The critical incident technique," *Psychological Bulletin*, vol. 51, no. 4, pp. 327-358, 1954, doi: 10.1037/h0061470.

[21] H. Beyer and K. Holtzblatt, *Contextual Design: Defining Customer-Centered Systems*. San Francisco, CA, USA: Morgan Kaufmann, 1998.

[22] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proceedings of the 40th International Conference on Machine Learning (ICML)*, 2023, pp. 28492-28518. [Online]. Available: https://proceedings.mlr.press/v202/radford23a.html

[23] OpenAI, "Whisper model card," GitHub. [Online]. Available: https://github.com/openai/whisper/blob/main/model-card.md. [Accessed: Jul. 26, 2026].

[24] M. Turner, B. Kitchenham, P. Brereton, S. Charters, and D. Budgen, "Does the technology acceptance model predict actual use? A systematic literature review," *Information and Software Technology*, vol. 52, no. 5, pp. 463-479, 2010, doi: 10.1016/j.infsof.2009.11.005.

[25] Lovable, "Terms and conditions," lovable.dev. [Online]. Available: https://lovable.dev/terms. [Accessed: Jun. 10, 2026].

[26] S. K. Karttunen, “Copyright of AI-generated works: Approaches in the EU and beyond,” European Parliamentary Research Service, European Parliament, Briefing PE 782.585, Dec. 2025. [Online]. Available: https://www.europarl.europa.eu/RegData/etudes/BRIE/2025/782585/EPRS_BRI(2025)782585_EN.pdf. [Accessed: Jun. 10, 2026].

Appendix

[A] Prototyping prompts

The following prompts were issued to the Lovable platform, in sequence, to generate and refine the high-fidelity prototype, and are reproduced verbatim. The initial build specification was drafted with the assistance of an AI language model, which expanded the authors’ description of the core elements into the detailed prompt below, including its layout and presentation conventions; the authors reviewed it before use. Because the platform is generative and non-deterministic, these prompts document the instructions given rather than guaranteeing an identical reconstruction.

```
Build a high-fidelity React-based Emergency Department Dashboard.

The goal is maximum situation awareness with zero scrolling (fit to screen height).
Use Tailwind CSS.

Layout & Navigation

Main background: Light Gray / Medical White.

Top Header: "Team Selection" Dropdown (Right: Overall, Team A, Team B, Team C, Team
X).

NEW: Team Status Bar: Below the header, create a horizontal strip containing 4 small
status cards (Team A, B, C, X). Each card shows: Team Name, a Status Dot
(Green/Yellow/Red), and Current Count. This gives a "glance" of all teams even when
"Overall ED" is selected.

Current Status Section (Grid Layout)

Create a 3-column layout for the current status. All charts should update based on the
team selection using hardcoded dummy data.
```

Total Patients Card: Large bold number. Below the number, show "Baseline: [X]" and a color-coded delta (+/-).

Patients by Age Span (Bar Chart): Vertical bars for 0-18, 19-65, 65+.

SPECIFIC REQUIREMENT: Instead of one global baseline line, draw a short, horizontal DASHED line over EACH individual bar representing the specific historical average for that age group.

Patients by Priority (Bar Chart): Standard Triage colors (Red, Orange, Yellow, Green, Blue).

SPECIFIC REQUIREMENT: Each bar must have its own individual DASHED horizontal baseline indicator at its specific historical average height.

Patient Flow Trends (Trend Comparison)

Create two side-by-side line charts for "Inflow" and "Outflow".

Dynamic Baselines: These charts MUST have TWO lines.

A solid, thick line (Primary color) for the "Current Actual Flow".

A dashed, thinner gray line (Secondary) that follows a CURVED path representing the "Historical Average Trend" for this time of day. Make the average trend extend past current time to display how the near future can look like based on "historic trends"

Timeframes: Add a toggle for "1h / 2h / 3h / Day / Week / Month" that smoothly updates the chart data.

X-axis: Time (HH:mm). Y-axis: Patient Count.

UI/UX Principles

Use Lucide-react for icons.

Ensure "No-Scroll" logic: Use "h-screen" and flex-col to make sure all elements fit on one screen.

Professional medical aesthetic: High contrast, clean typography (Sans-serif), and clear borders between modules.

For reference: in a day about 80-120 patients get taken care of; 560-840 a week; 2240-3360 a month

Use shadcn/ui components or similar for the dropdowns and cards for a polished look.

"Update the patient distribution across the four teams to reflect realistic clinical allocation patterns:

Team A:

Age distribution should skew older, majority of patients between 65–90 years old

Mix of triage priorities, mostly yellow and orange with a few red

Represents the typical high-complexity older patient cohort

Team B:

Age distribution should be predominantly young patients, majority between 5–35 years old

Mostly green and yellow triage priorities

Very few or no patients above 60 years old

Team C:

Junior staff team so patients should be more stable and straightforward

Majority green triage priority, some yellow, no red patients

Age distribution can be mixed but avoid very old or very critical patients

Lower average waiting times to reflect simpler cases

Team X:

Age distribution slightly older, with a notable portion of patients between 55–80 years old

Higher proportion of orange and red triage patients compared to all other teams

This team handles the most acute and complex cases

Average wait times should reflect the higher complexity

Make sure the triage color distribution and age values in the patient list rows actually reflect these rules — not just in summary charts but in the individual patient entries as well."

"Make the following three UI fixes to the dashboard:

Fix 1 — Inflow and Outflow font color: The numerical totals displayed on the Inflow and Outflow cards should use the same colors as their corresponding graph lines. The inflow number should be the same blue as the inflow graph line. The outflow number should be the same green as the outflow graph line. Currently the numbers appear in a default/black font which does not match.

Fix 2 — Average baseline visibility on Patient Load: Add a clearly visible background band or shaded zone around the average line on the Patient Load graph. The band should make it immediately obvious whether the current load is above or below average — for

example a light red shaded area above the average line and a light green shaded area below it. The average line itself should remain visible on top of this shading.

Fix 3 — Time period selector placement: Move the 1h / 2h / 3h / Day / Week / Month toggle buttons from their current position to the top right of the dashboard, placed directly below the team selection tabs. It should sit flush to the right side and be clearly grouped as a ti

---

"Fix the average indicator text label on the Patient Load graph. Do not add any background shading or coloring to the graph itself.

Instead, the text label that displays the average value should have a small colored pill or badge box directly behind it — similar to a highlighted tag. If the current load is above average the box behind the text should be red. If the current load is below average the box behind the text should be green. The text itself should remain readable on top of the colored box. The graph background stays clean and unchanged."

"Make the following two UI changes:

Fix 1 — Remove dropdown and add Overall button: Remove the dropdown menu for team selection entirely. Instead add an 'Overall' option directly as a button alongside the existing Team A, Team B, Team C and Team X buttons so all five options sit in the same button row. Overall should be the default selected state and show aggregated data across all teams.

Fix 2 — Patient Load text visibility: Increase the visibility of the text inside the Patient Load component. Make the font larger, bolder or increase the contrast against the background — whichever is needed so the numbers and labels are clearly readable at a glance without having to look closely."

Add an "Overall ED" button to the left of the team selection bar (before Team A). It should match the same style as the team buttons but act as a view toggle — when selected, it highlights as active (e.g. bold border or filled background) and the dashboard shows the overall ED data. The other team buttons should deselect when Overall ED is active, and vice versa. Use the current active state styling from the existing team buttons as a reference.